DEPRESSION AND SUICIDALITY

A COMPREHENSIVE EXAMINATION OF MECHANISTIC FRAMEWORKS

Dawn Jensen, PhD

Overview

Overview

All Models are Wrong, Some Models are Useful

Overview

All Models are Wrong, Some Models are Useful

Neurobiological Perspectives

Depression is a leading cause of disability worldwide

Depression is a leading cause of disability worldwide

~6% wordwide experience a depressive disorder

~36% of disability attributed to mental health disorders

Depression is a leading cause of disability worldwide

Suicide is one of the leading causes of preventable death

Depression is a leading cause of disability worldwide

Suicide is one of the leading causes of preventable death

Responsible for ~49k deaths
14.3 million had serious thoughts about suicide
4.6 million planned an attempt
2.2 million attempted suicide
In 2024

Depression is a leading cause of disability worldwide

Suicide is one of the leading causes of preventable death

Predicting who will develop debilitating depression or attempt suicide remains difficult, despite a large body of knowedge regarding risk factors

What is Depression?

Primary DSM-V depression disorders, criteria for adults

Depressive Diagnosis	Symptoms
Major Depressive Episode	
• 5 or more of the 9 DSM-5 major depression symptoms to be present within a 2-week period and represent a change from previous functioning • At least one of the symptoms is either (1) depressed mood or (2) loss of interest or pleasure • Cause clinically significant distress or impairment in social, occupational, or other important areas of functioning • Episode is not attributable to the physiological effects of a substance or to another medical condition • There has never been a manic episode or a hypomanic episode	• Depressed most of the day, nearly every day • Markedly diminished interest or pleasure in all, or almost all, activities most of the day, nearly every day • Significant weight loss when not dieting or weight gain • Insomnia or hypersomnia nearly every day • Psychomotor agitation or retardation nearly every day • Fatigue or loss of energy nearly every day • Feelings of worthlessness or excessive or inappropriate guilt nearly every day • Diminished ability to think or concentrate, or indecisiveness, nearly every day • Recurrent thoughts of death, recurrent suicidal ideation without a specific plan, or a suicide attempt or a specific plan for committing suicide

CORE DIMENSIONS OF DEPRESSION

Negative Affect

CORE DIMENSIONS OF DEPRESSION

Negative Affect

Anhedonia

CORE DIMENSIONS OF DEPRESSION

Negative Affect

Anhedonia

Rumination

CORE DIMENSIONS OF DEPRESSION

Negative Affect

Anhedonia

Rumination

Cognitive Slowing

CORE DIMENSIONS OF DEPRESSION

Negative Affect

Anhedonia

Rumination

Cognitive Slowing

Social Withdrawl

CORE DIMENSIONS OF DEPRESSION

Negative Affect

Anhedonia

Rumination

Cognitive Slowing

Social Withdrawl

Irritability

What is Suicidality?

Review of the DSM-5-TR Revisions

Disorder	Revisions in the DSM-5-TR	Relevant MHS Asessment	Impact
Austism Spectrum Disorder	Criterion A was clarified to indicate that all symptoms are required.	• Autism Spectrum Rating Scales™ (ASRS®) • Conners Comprehensive Behavior Rating Scales™ (Conners CBRS®)	No impact on test items, test scores, or test interpretation as the ASRS and Conners CBRS.
Bipolar and Related Disorders	The criteria and specifiers for Bipolar and Related Disorders have been edited for clarification, including changes to facilitate differential diagnostic decisions regarding psychotic disorders.	• Conners Comprehensive Behavior Rating Scales™ (Conners CBRS®)	No impact on test items, test scores, or test interpretation on Conners CBRS.
Depressive Disorders	The criteria for Depressive Disorders have been edited for clarification, including changes to facilitate differential diagnostic decisions regarding psychotic disorders.	• Children's Depression Inventory 2nd Edition (CDI 2®) • Conners Comprehensive Behavior Rating Scales™ (Conners CBRS®)	No impact on test items, test scores, or test interpretation for the CDI 2 and Conners CBRS.
Post-Traumatic Stress Disorder	A redundant description under Criterion A for children was removed.	• Conners Comprehensive Behavior Rating Scales™ (Conners CBRS®)	No impact on test items, test scores, or test interpretation on existing screener items.
Suicidal Behavior and Non-suicidal Self-Injury	Codes added to *Other Conditions That May Be a Focus of Clinical Attention.*	• Conners 4th Edition (Coming soon) • Conners Comprehensive Behavior Rating Scales™ (Conners CBRS®)	No impact on test items, test scores, or test interpretation. Items on the Conners 4 and Conners CBRS can be used to help screen for suicidal behavior and non-suicidal self-injury.

Symptoms of Suicidal Behavior Disorder

According to the DSM-5, there are five proposed criterion Suicidal Behavior Disorder, with two specifiers

1. The individual has made a suicide attempt within the past two years.
2. The criterion for non-suicidal self-injurious behavior is not met during the aforementioned suicide attempts.
3. The diagnosis is not applied to preparation for a suicide attempt, or suicidal ideation.
4. the act was not attempted during an altered mental state, such as delirium or " confusion".
5. The act was not ideologically motivated- e.g. - religious or political.

Other specifiers are:

- Current- Not more than 12- 24 months since last attempt.
- In Remission- more than 24 months since last attempt. (American Psychiatric Association, 2013).

CORE DIMENSIONS OF SUICIDALITY

Passive Death Wishes

CORE DIMENSIONS OF SUICIDALITY

Passive Death Wishes

Suicidal Ideation

CORE DIMENSIONS OF SUICIDALITY

Passive Death Wishes

Suicidal Ideation

Planning

CORE DIMENSIONS OF SUICIDALITY

Passive Death Wishes

Suicidal Ideation

Planning

Attempts

CORE DIMENSIONS OF SUICIDALITY

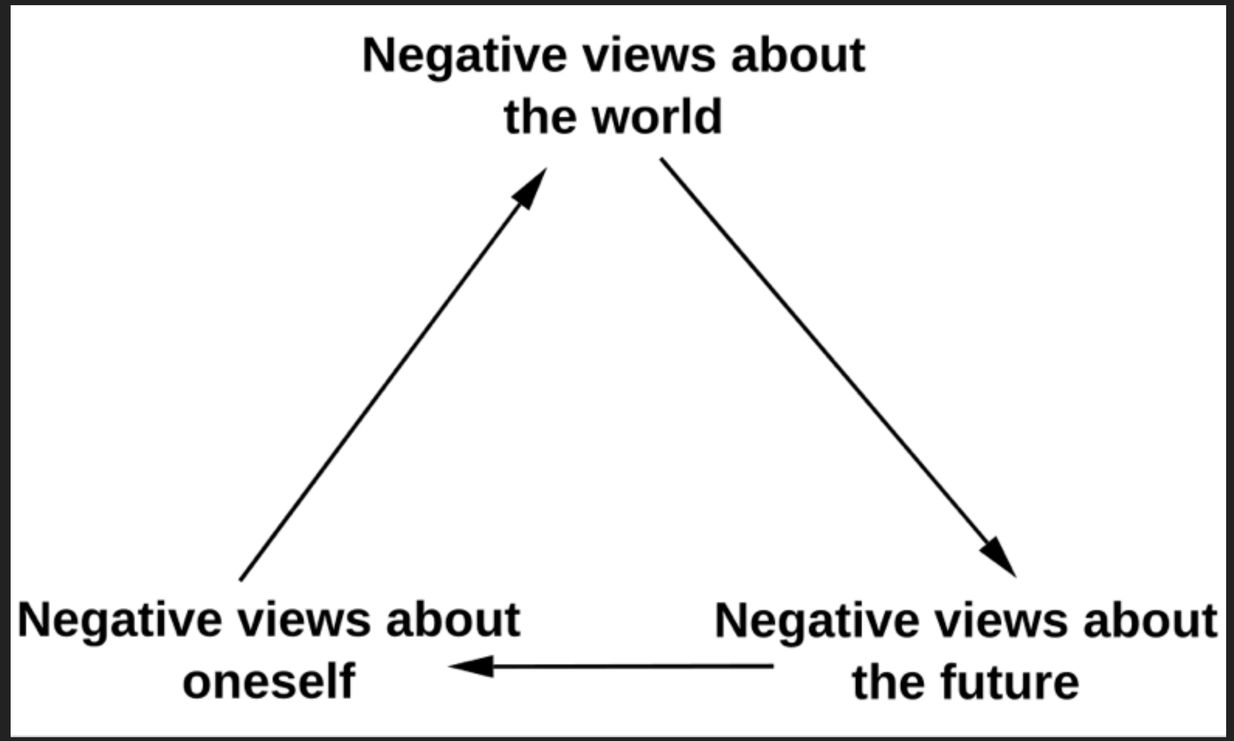
The mechanisms producing suicidal thoughts may differ from those producing suicidal actions

Cognitive Model of Depression

All Models are Wrong, Some Models are Useful

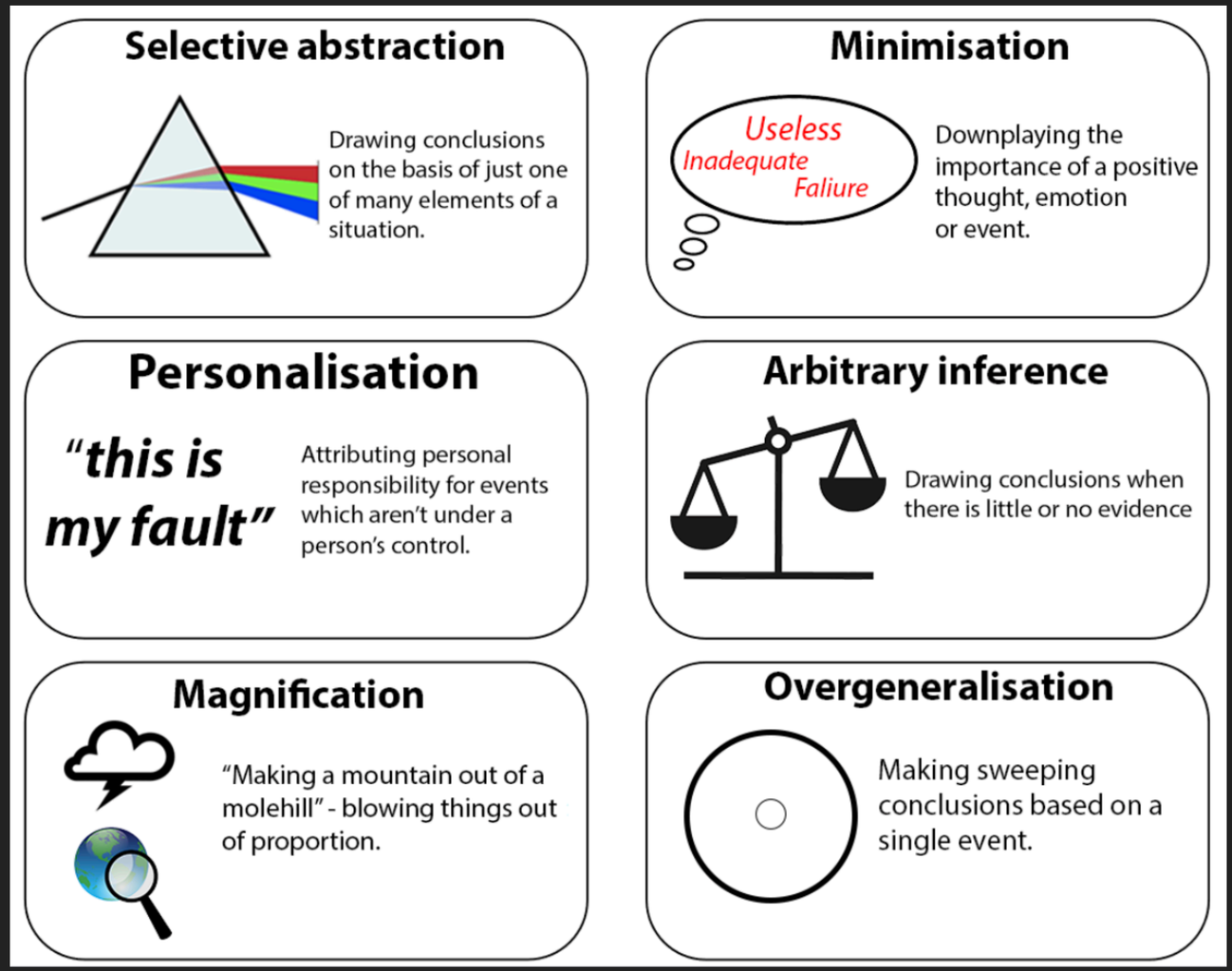
THE NEGATIVE TRIAD

- Depressive disorders are characterized by dysfunctional negative views of themselves, their life experience, and their future
- The self - "I'm worthless", "I wish I was different"
- The world - "No one values me", "People ignore me all the time"
- The future - "I'm hopeless because things will never change", "things can only get worse"



COGNITIVE DISTORTIONS

- A type of cognitive bias of maladaptive compulsive thought patterns
- Create a foundation for a negative self-schema that minimizes positive experiences and achievements
- Eventually developed the Beck Depression Inventory



PROBLEMS WITH THE COGNITIVE MODEL

Causality Paradox:

Do negative thoughts cause depression or does depression cause negative thoughts?

PROBLEMS WITH THE COGNITIVE MODEL

Causality Paradox

Avoids Emotional Root Causes:

Treatment (Cognitive Behavioral Therapy) addresses symptoms, but does not explore underlying causes

PROBLEMS WITH THE COGNITIVE MODEL

Causality Paradox

Avoids Emotional Root Causes

Fails to Explain More Severe Symptoms:

Rumination, hallucinations, comorbidities, persistance

PROBLEMS WITH THE COGNITIVE MODEL

Causality Paradox

Avoids Emotional Root Causes

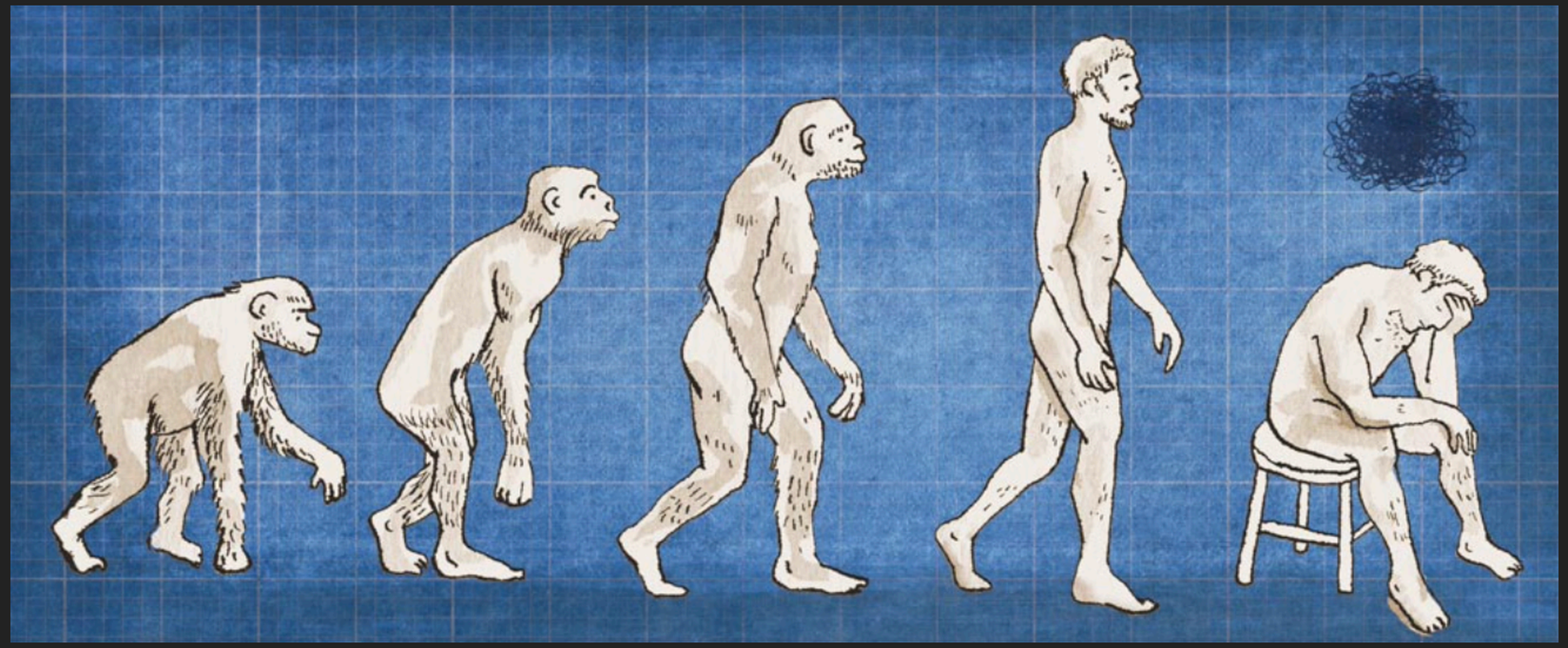
Fails to Explain More Severe Symptoms

Fails to Account for Susceptibility:

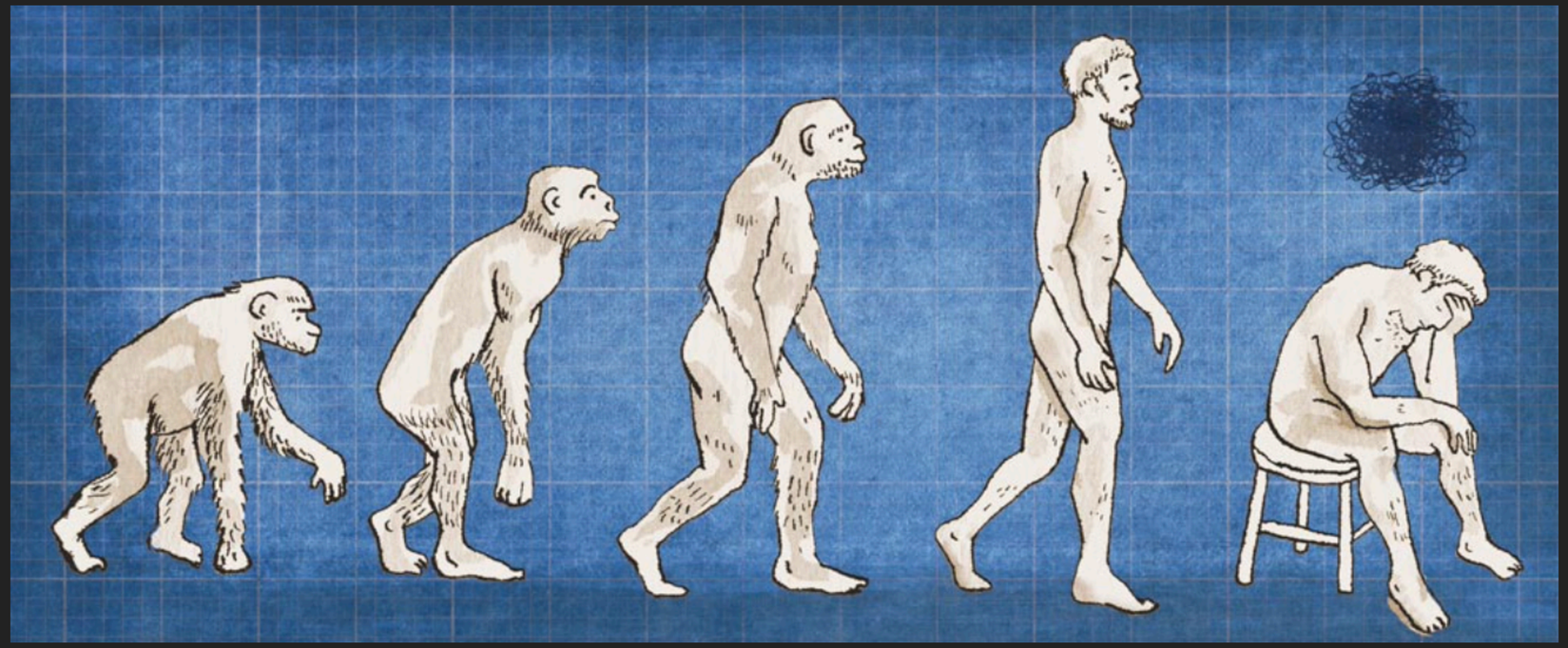
Cannot account for the interplay between genetic, enviroment, and ptsd

Biological Adaptation Model of Depression

All Models are Wrong, Some Models are Useful



Analytical Rumination Hypothesis: Some depressive symptoms may have evolved to facilitate analysis of difficult social problems



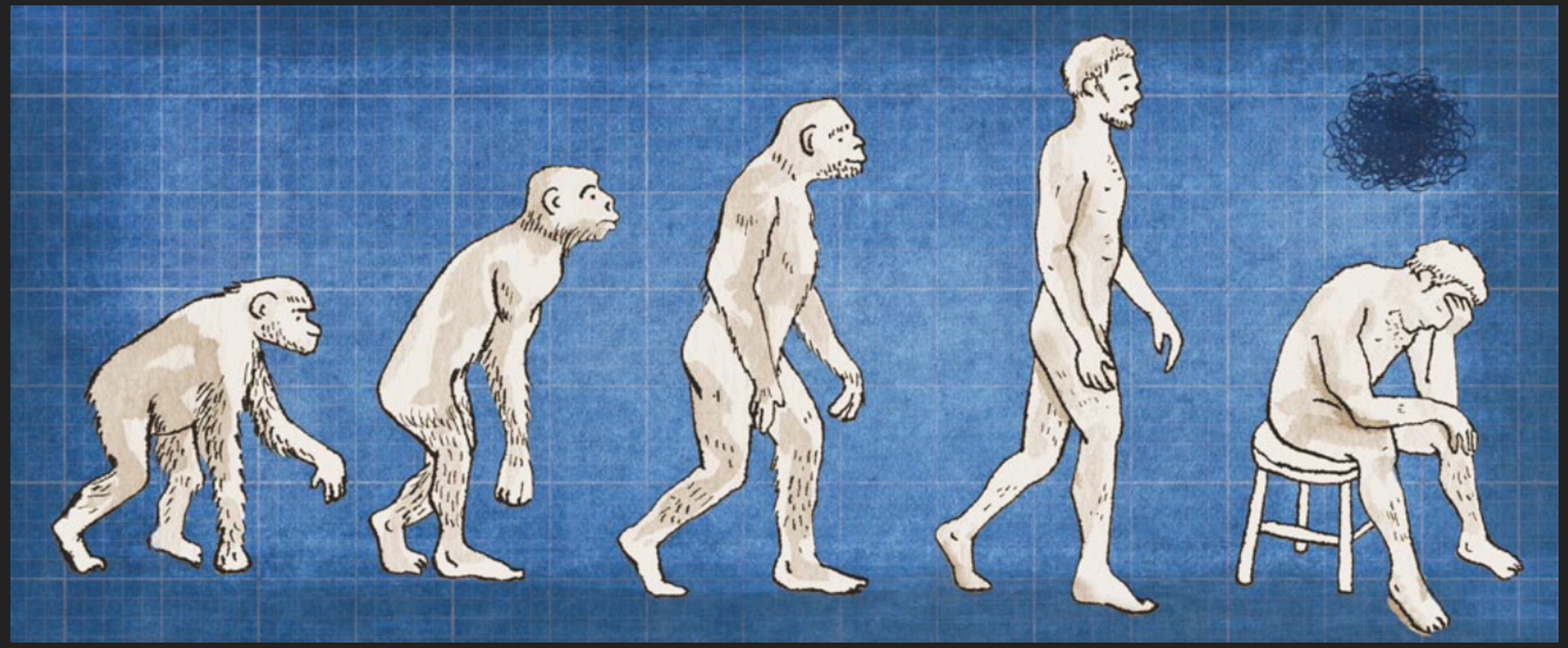
Analytical Rumination Hypothesis

HYPOTHESIZED BENEFITS

Focused attention

Reduced distraction through isolation

Sustained problem analysis



Analytical Rumination Hypothesis

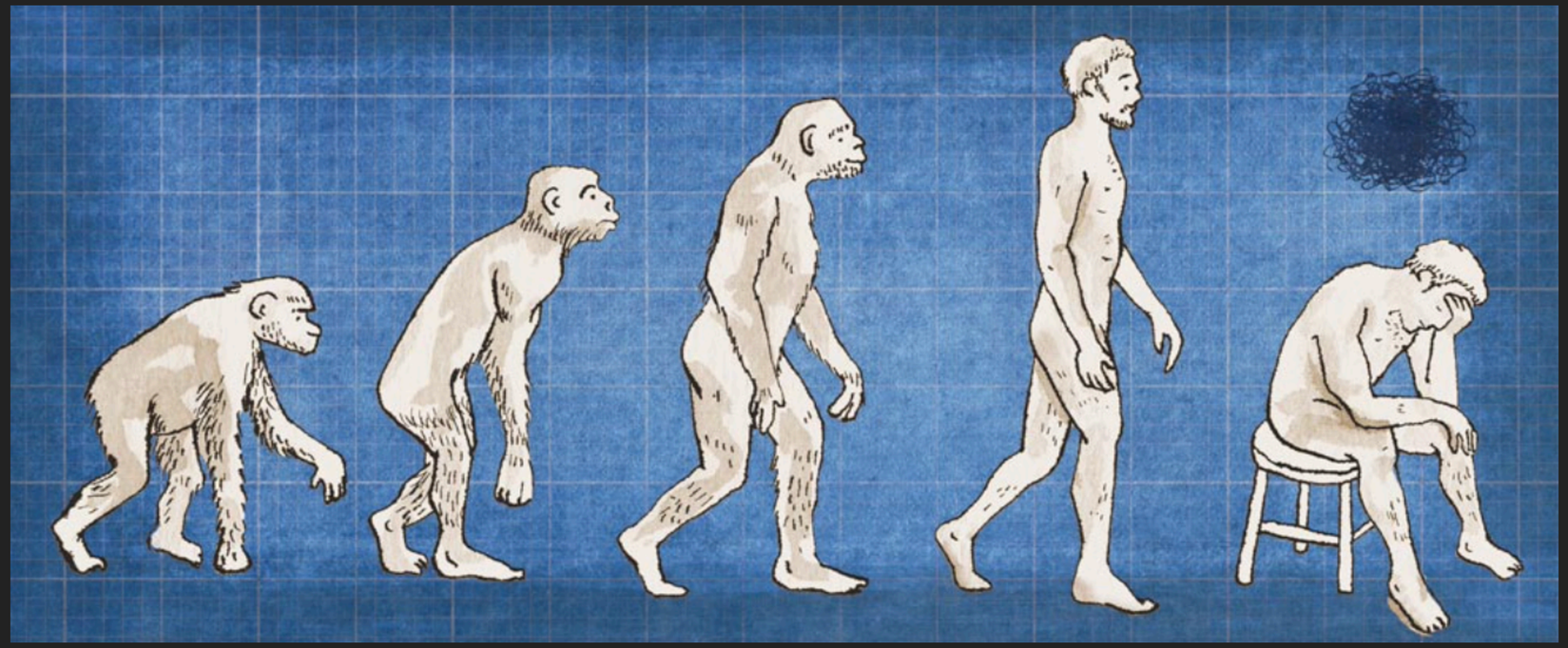
CRITIQUES

Severe depression is maladaptive

Cognitive impairment

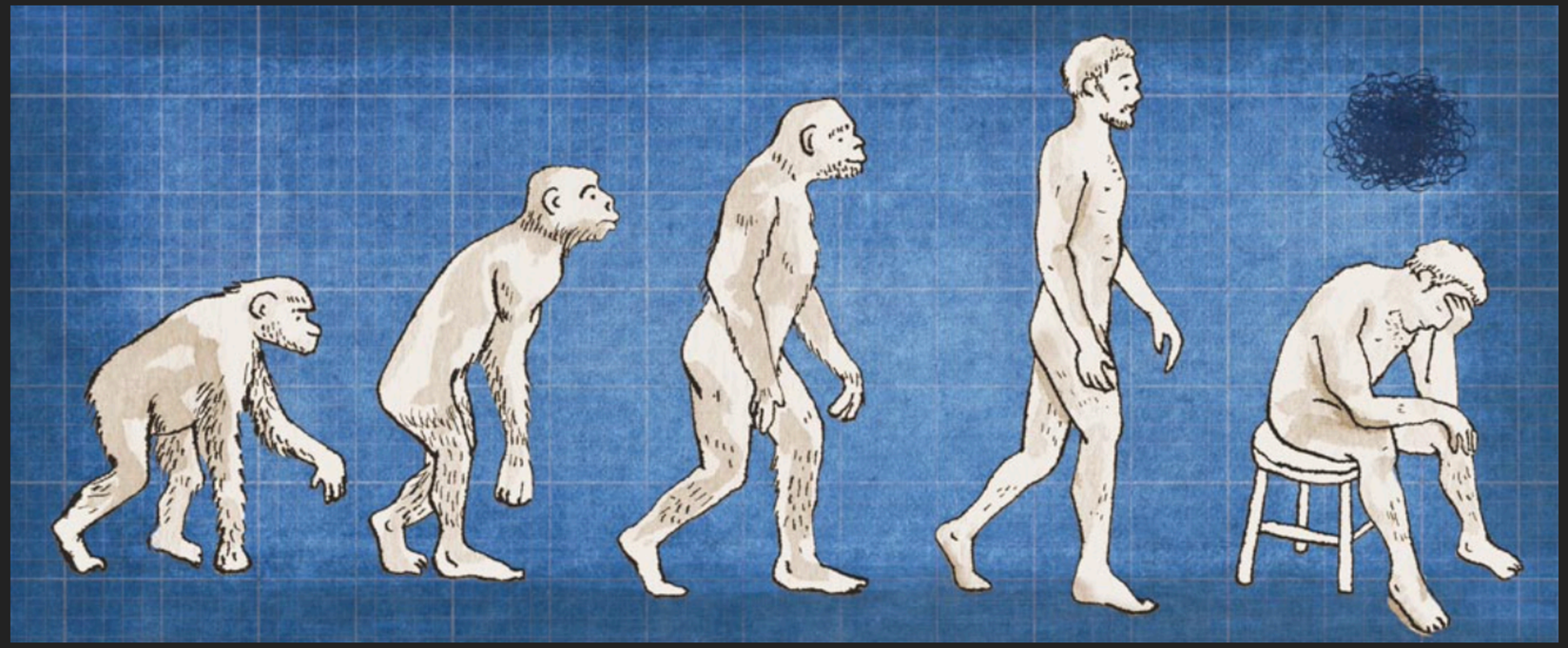
Reduced functioning

Increased mortality



Social Rank Models: Depression may represent an evolved response to defeat, loss of status, or failed competition

Withdrawl may historically have reduced risk of further harm



Social Rank Models: Depression may represent an evolved response to defeat, loss of status, or failed competition

Withdrawl may historically have reduced risk of further harm

As yet, there is no imperical evidence that supports this model

Chemical Imbalance Model of Depression

All Models are Wrong, Some Models are Useful

THE SEROTONIN HYPOTHESIS

Popular with the rise of Selective Serotonin Reuptake Inhibitors (SSRIs) in the 1990s.

THE SEROTONIN HYPOTHESIS

Popular with the rise of Selective Serotonin Reuptake Inhibitors (SSRIs) in the 1990s.

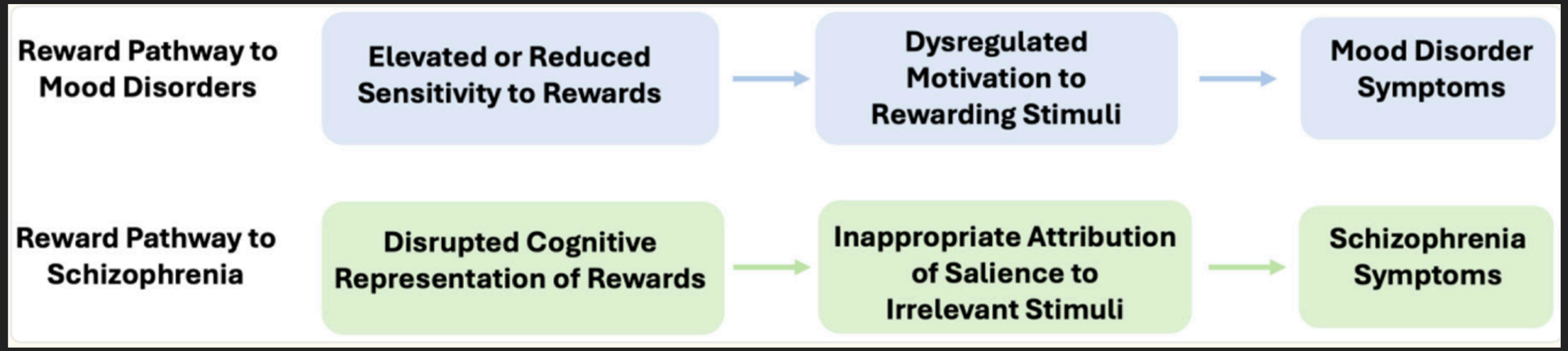
Centered around the theory that low serotonin caused depression and increasing the amount of time serotonin stayed in the synaptic cleft would cure the root biological defect

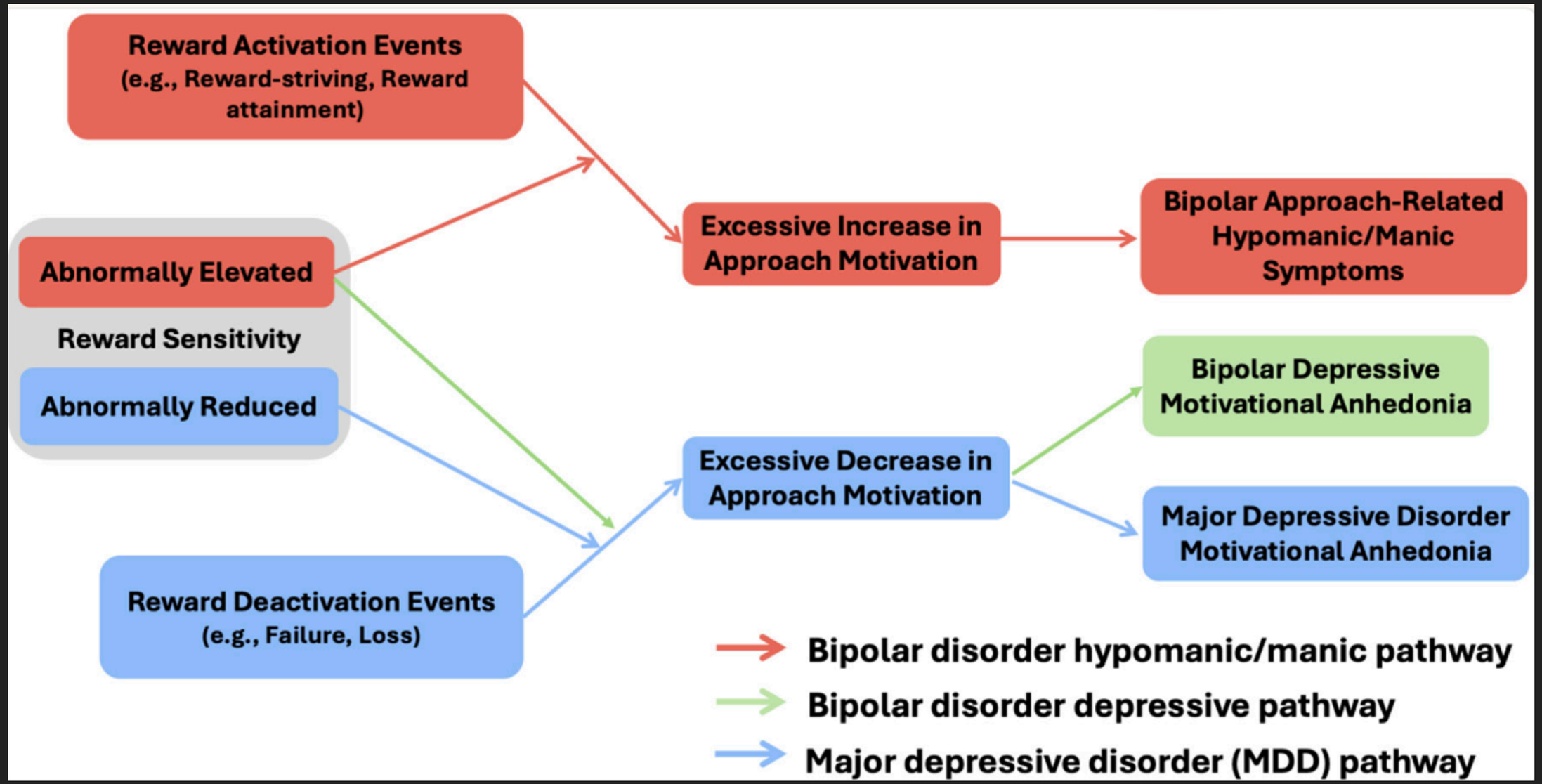
MOVING BEYOND SEROTONIN

Recent research no longer supports this as a sufficient explanation of depression. Evidence suggests a more complex, heterogeneious biology

Reward Dysfunction Model of Depression

All Models are Wrong, Some Models are Useful

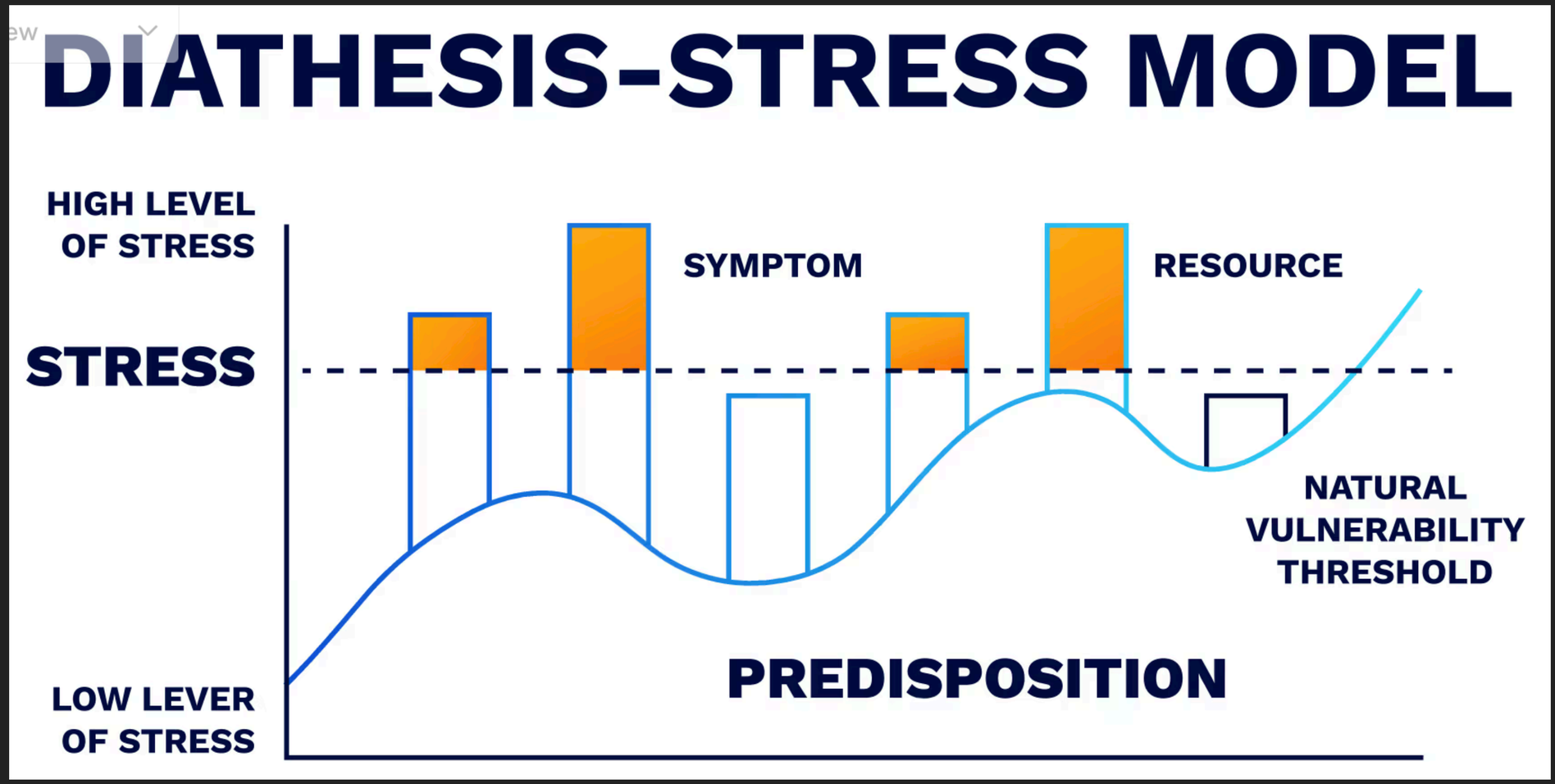

Reward Pathway to Mood Disorders
Elevated or Reduced Sensitivity to Rewards
Dysregulated Motivation to Rewarding Stimuli
Mood Disorder Symptoms
Reward Pathway to Schizophrenia
Disrupted Cognitive Representation of Rewards
Inappropriate Attribution of Salience to Irrelevant Stimuli
Schizophrenia Symptoms


Reward Activation Events
(e.g., Reward-striving, Reward attainment)
Abnormally Elevated
Reward Sensitivity
Abnormally Reduced
Reward Deactivation Events
(e.g., Failure, Loss)
Excessive Increase in Approach Motivation
Excessive Decrease in Approach Motivation
Bipolar Approach-Related Hypomanic/Manic Symptoms
Bipolar Depressive Motivational Anhedonia
Major Depressive Disorder Motivational Anhedonia
Bipolar disorder hypomanic/manic pathway
Bipolar disorder depressive pathway
Major depressive disorder (MDD) pathway

Stress-Diathesis Model of Depression

All Models are Wrong, Some Models are Useful

DEPRESSION EMERGES WHEN ENVIRONMENTAL STRESS EXCEEDS ADAPTIVE CAPACITY



DEPRESSION EMERGES WHEN ENVIRONMENTAL STRESS EXCEEDS ADAPTIVE CAPACITY

Risk accumulates through:

Genetics

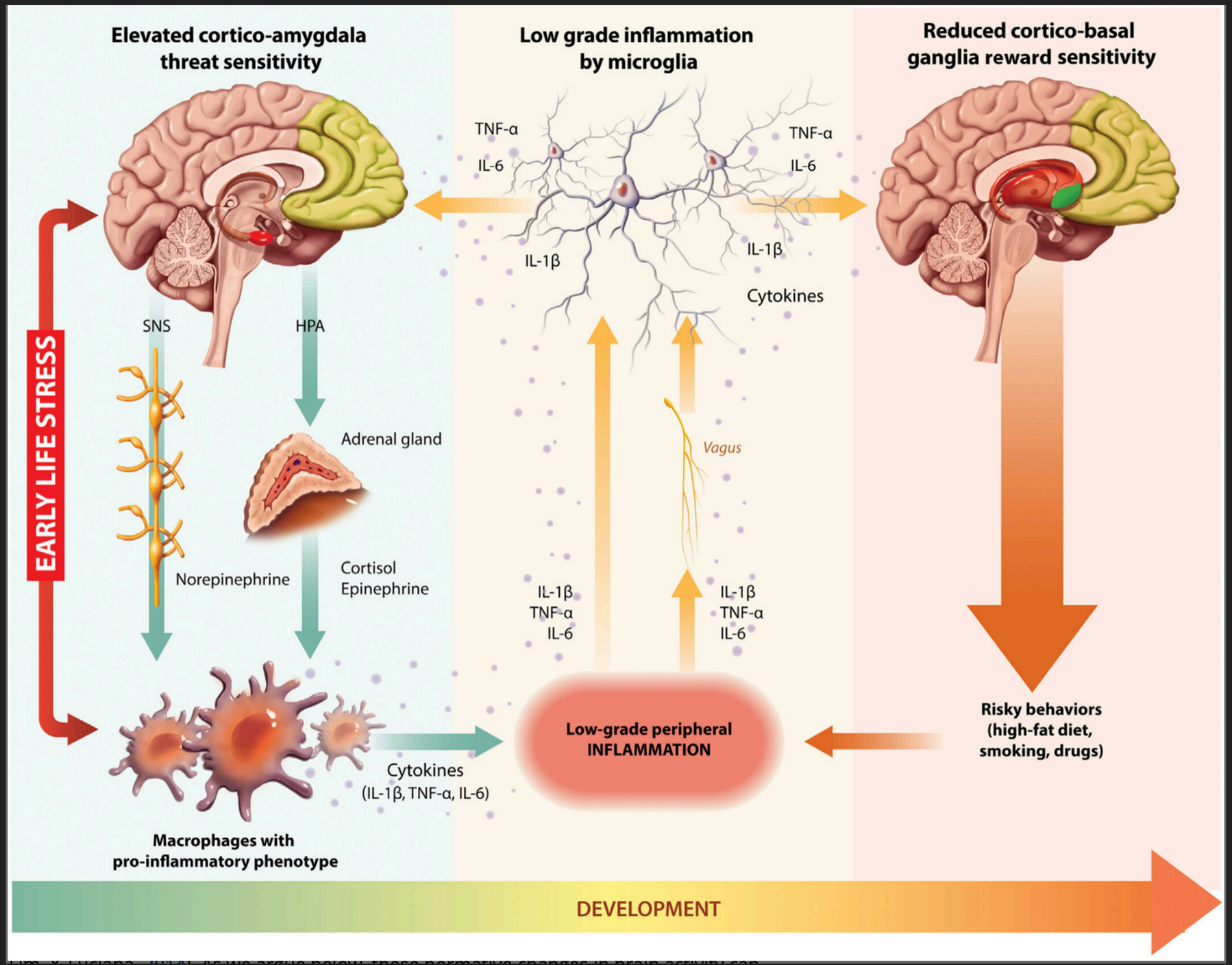
Adversity

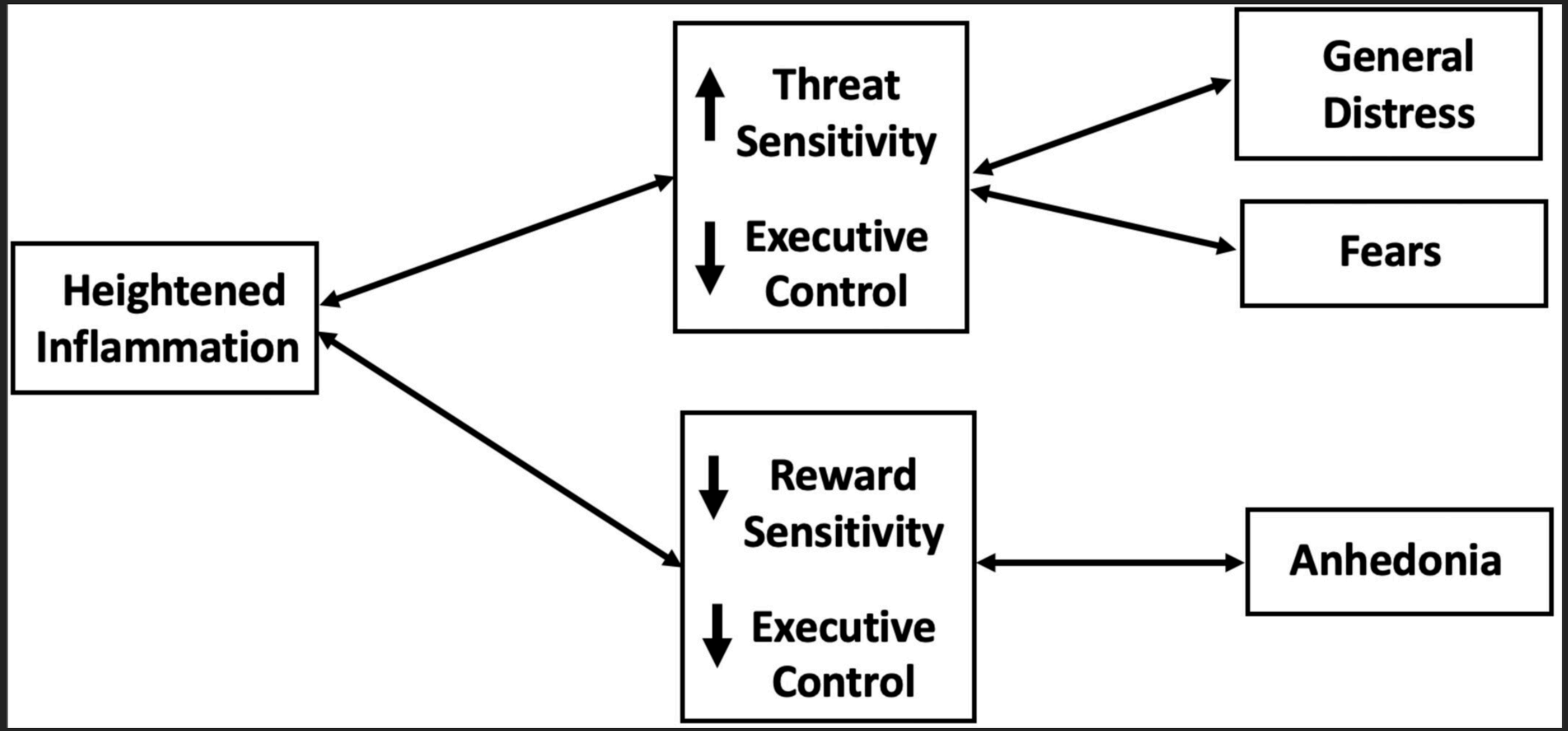
Cognitive Vulnerability

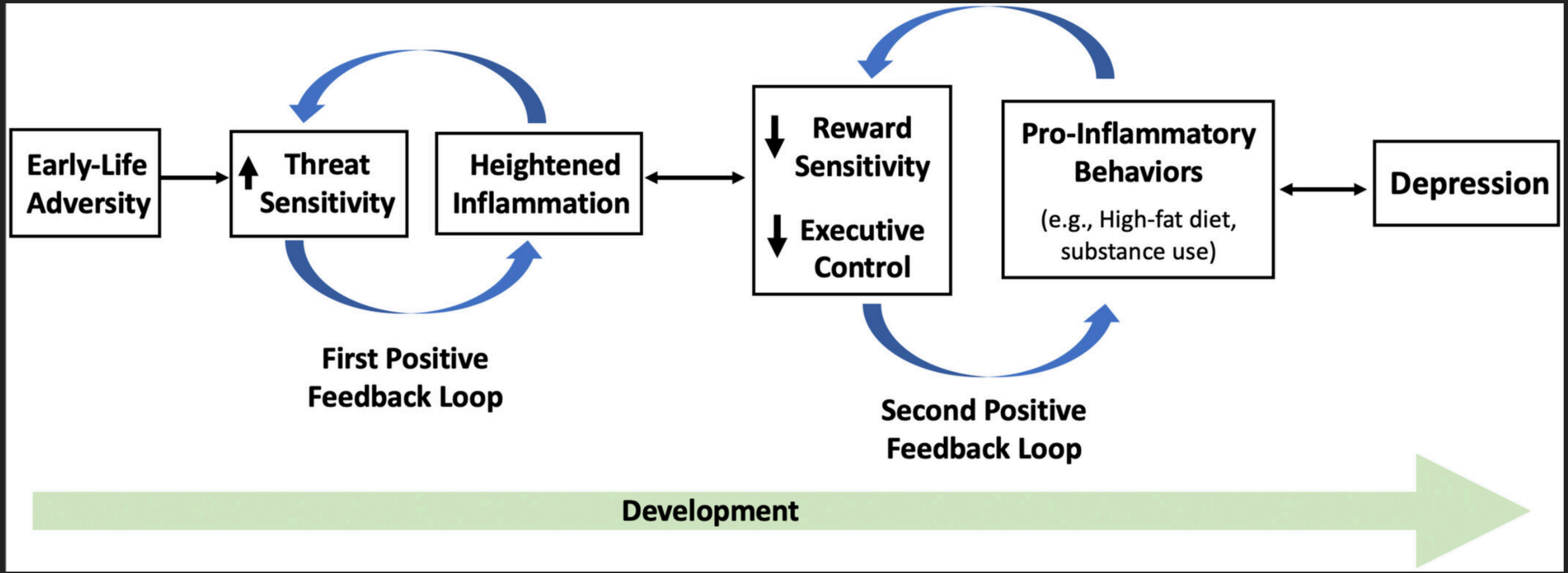
Social Stressors

Neuroinflammation Model of Depression

All Models are Wrong, Some Models are Useful

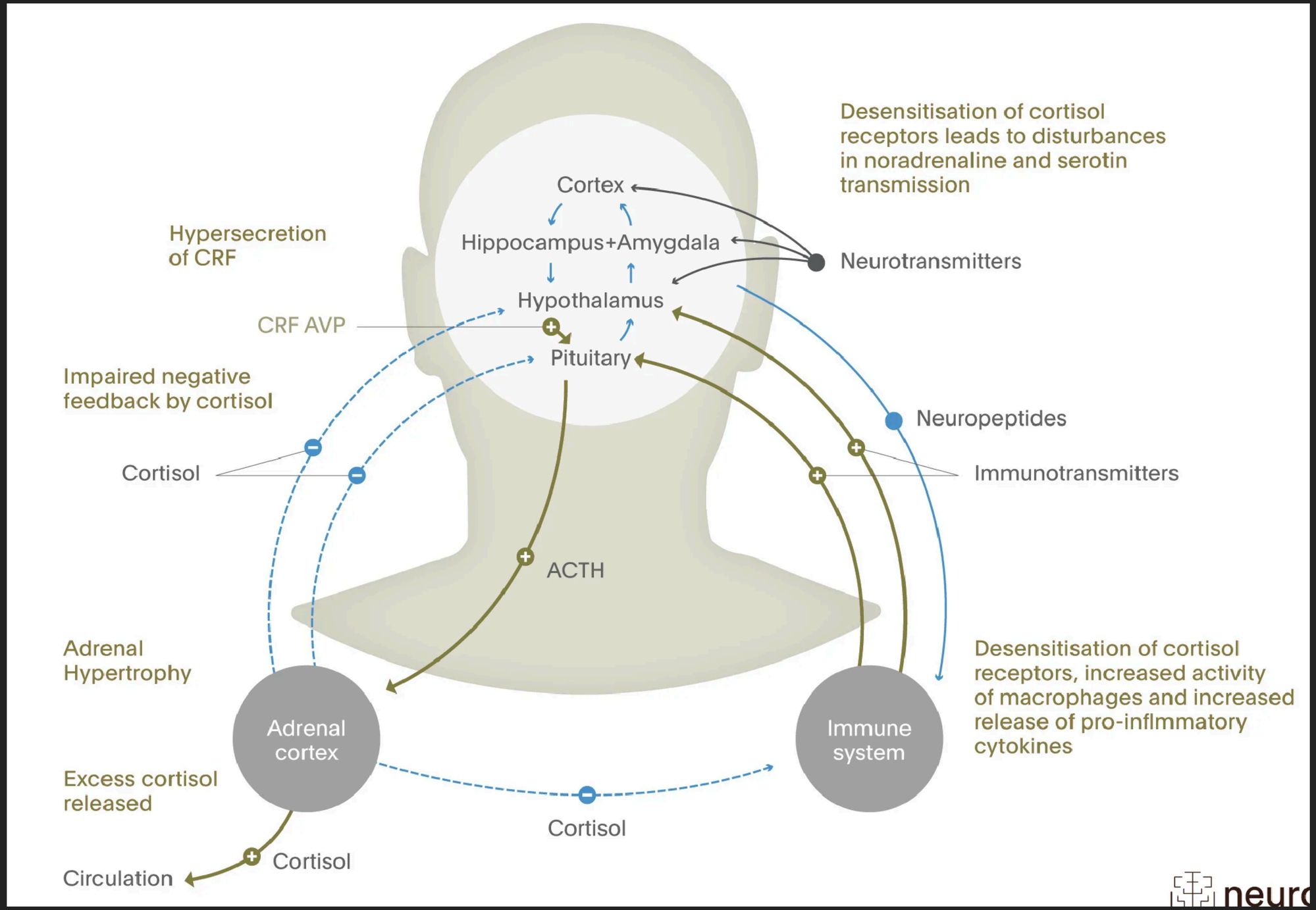

Elevated cortico-amygdala threat sensitivity
Low grade inflammation by microglia
Reduced cortico-basal ganglia reward sensitivity
TNF-α
IL-6
TNF-α
IL-6
IL-1β
IL-1β
Cytokines
EARLY LIFE STRESS
SNS
HPA
Adrenal gland
Vagus
Norepinephrine
Cortisol
Epinephrine
IL-1β
TNF-α
IL-6
IL-1β
TNF-α
IL-6
Low-grade peripheral INFLAMMATION
Risky behaviors (high-fat diet, smoking, drugs)
Cytokines
(IL-1β, TNF-α, IL-6)
Macrophages with pro-inflammatory phenotype
DEVELOPMENT


Heightened Inflammation
Threat Sensitivity
Executive Control
General Distress
Fears
Reward Sensitivity
Executive Control
Anhedonia


Early-Life Adversity
Threat Sensitivity
Heightened Inflammation
First Positive Feedback Loop
Reward Sensitivity
Executive Control
Pro-Inflammatory Behaviors
(e.g., High-fat diet, substance use)
Depression
Second Positive Feedback Loop
Development

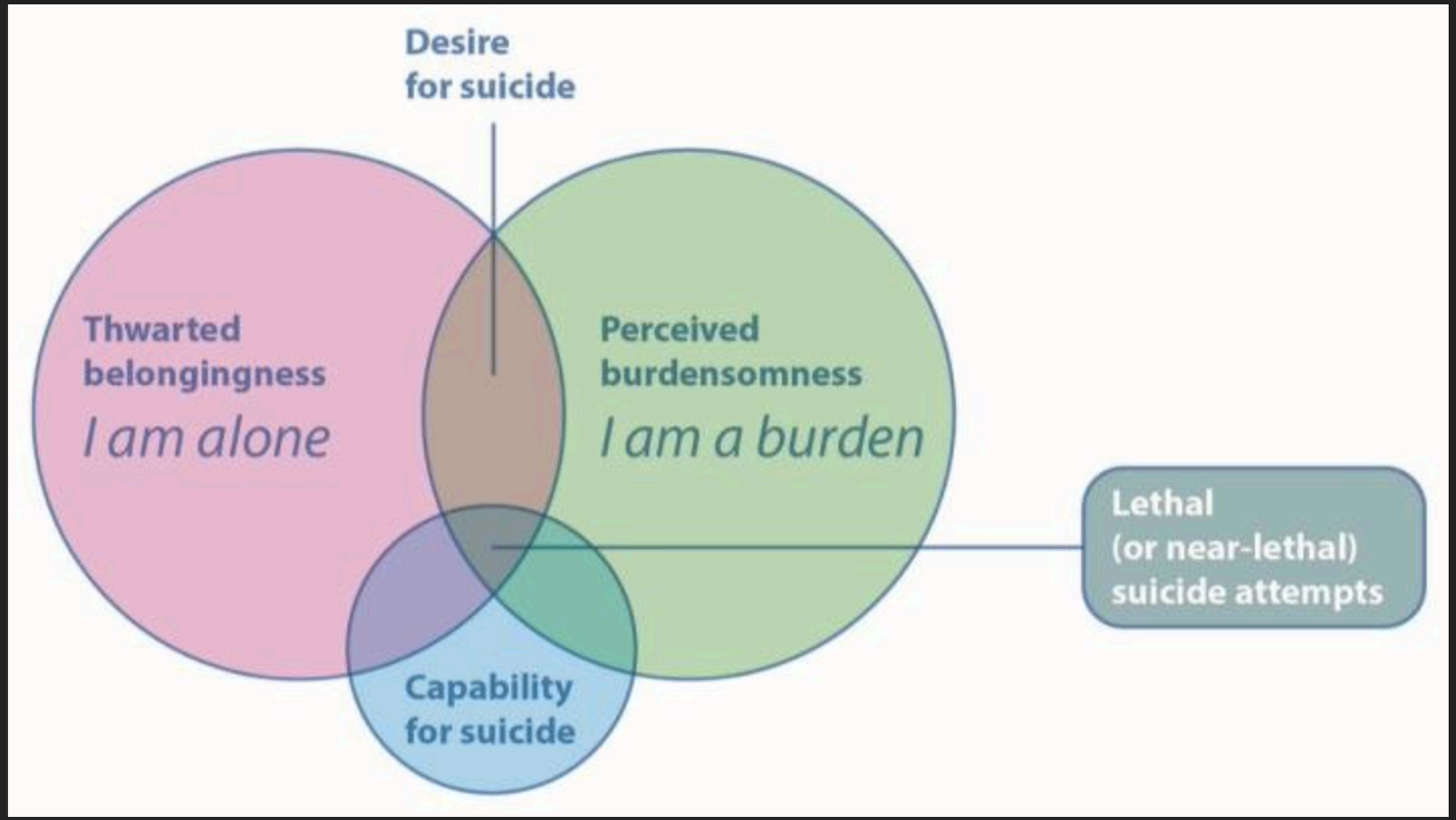
Hypothalamus Pituitary Adrenal Axis Model of Depression

All Models are Wrong, Some Models are Useful


Desensitisation of cortisol receptors leads to disturbances in noradrenaline and serotin transmission
Cortex
Hippocampus+Amygdala
Neurotransmitters
Hypersecretion of CRF
Hypothalamus
CRF AVP
Pituitary
Impaired negative feedback by cortisol
Neuropeptides
Cortisol
Immunotransmitters
ACTH
Adrenal Hypertrophy
Desensitisation of cortisol receptors, increased activity of macrophages and increased release of pro-inflmmatory cytokines
Adrenal cortex
Immune system
Excess cortisol released
Cortisol
Cortisol
Circulation

Interpersonal Theory of Suicide

All Models are Wrong, Some Models are Useful


Desire
for suicide
Thwarted
belongingness
I am alone
Perceived
burdensomness
I am a burden
Capability
for suicide
Lethal
(or near-lethal)
suicide attempts

Integrated Motivatinal-Volitinal Model of Suicide

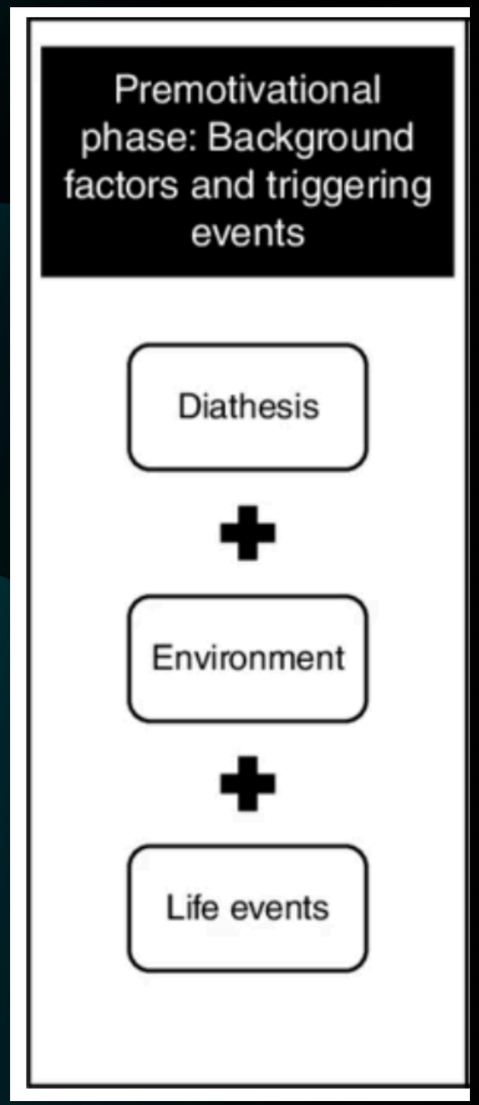
All Models are Wrong, Some Models are Useful

INTEGRATED MOTIVATINAL-VOLITINAL MODEL OF SUICIDE

Three Phases:

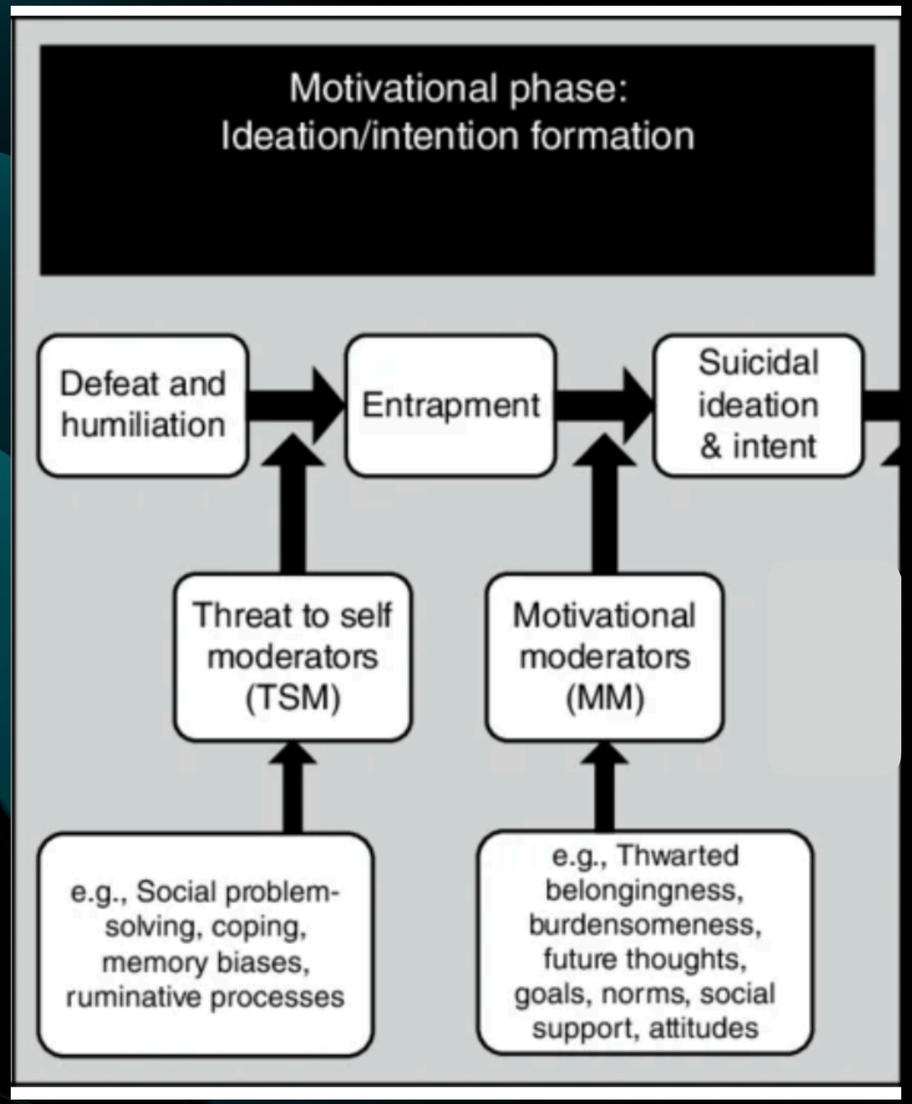
INTEGRATED MOTIVATINAL-VOLITIONAL MODEL OF SUICIDE

Three Phases:



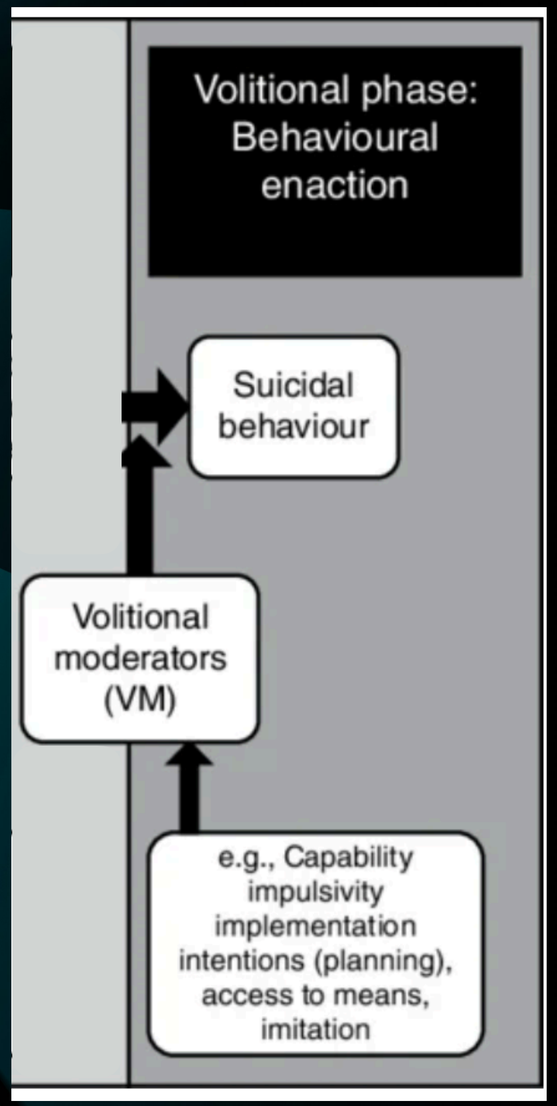
INTEGRATED MOTIVATINAL-VOLITIONAL MODEL OF SUICIDE

Three Phases:



INTEGRATED MOTIVATINAL-VOLITIONAL MODEL OF SUICIDE

Three Phases:



Neuroimaging Findings in Depression

Neurobiological Perspectives

From: Instability of default mode network connectivity in major depression: a two-sample confirmation study

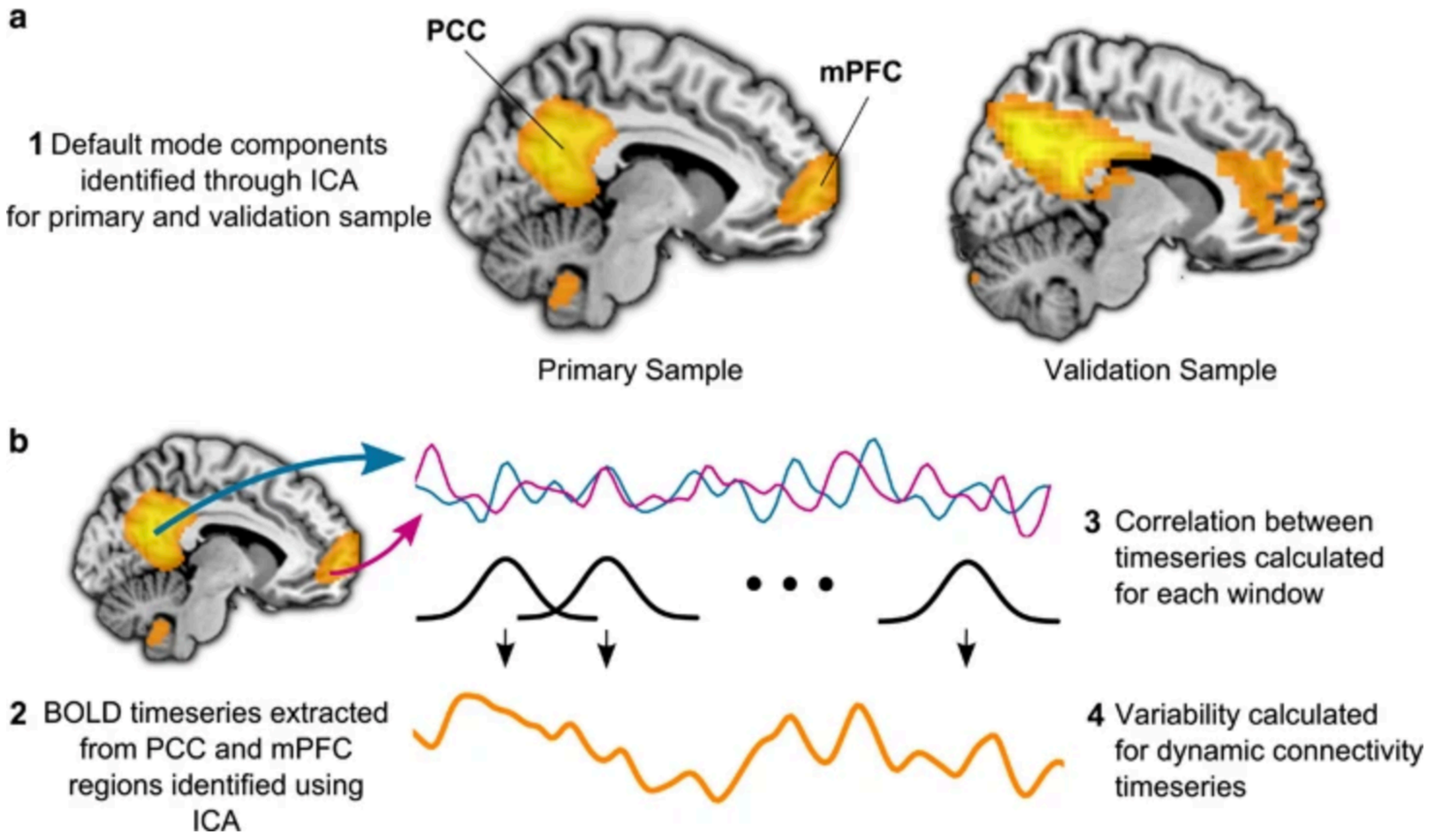


Fig. 1: Frontostriatal salience network is expanded nearly twofold in the cortex of highly sampled individuals with depression.

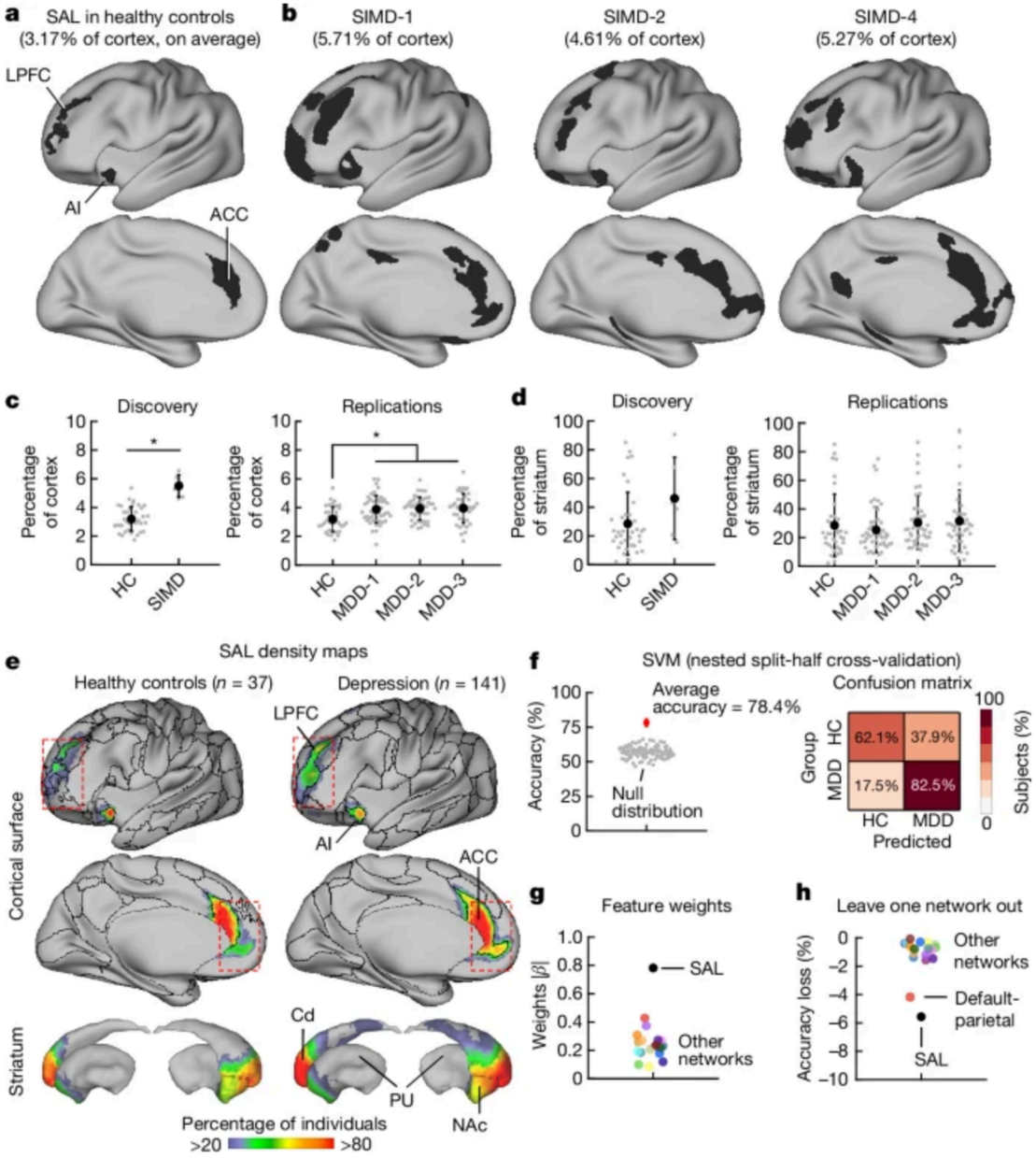


Fig. 2: Three modes of salience network expansion in depression.

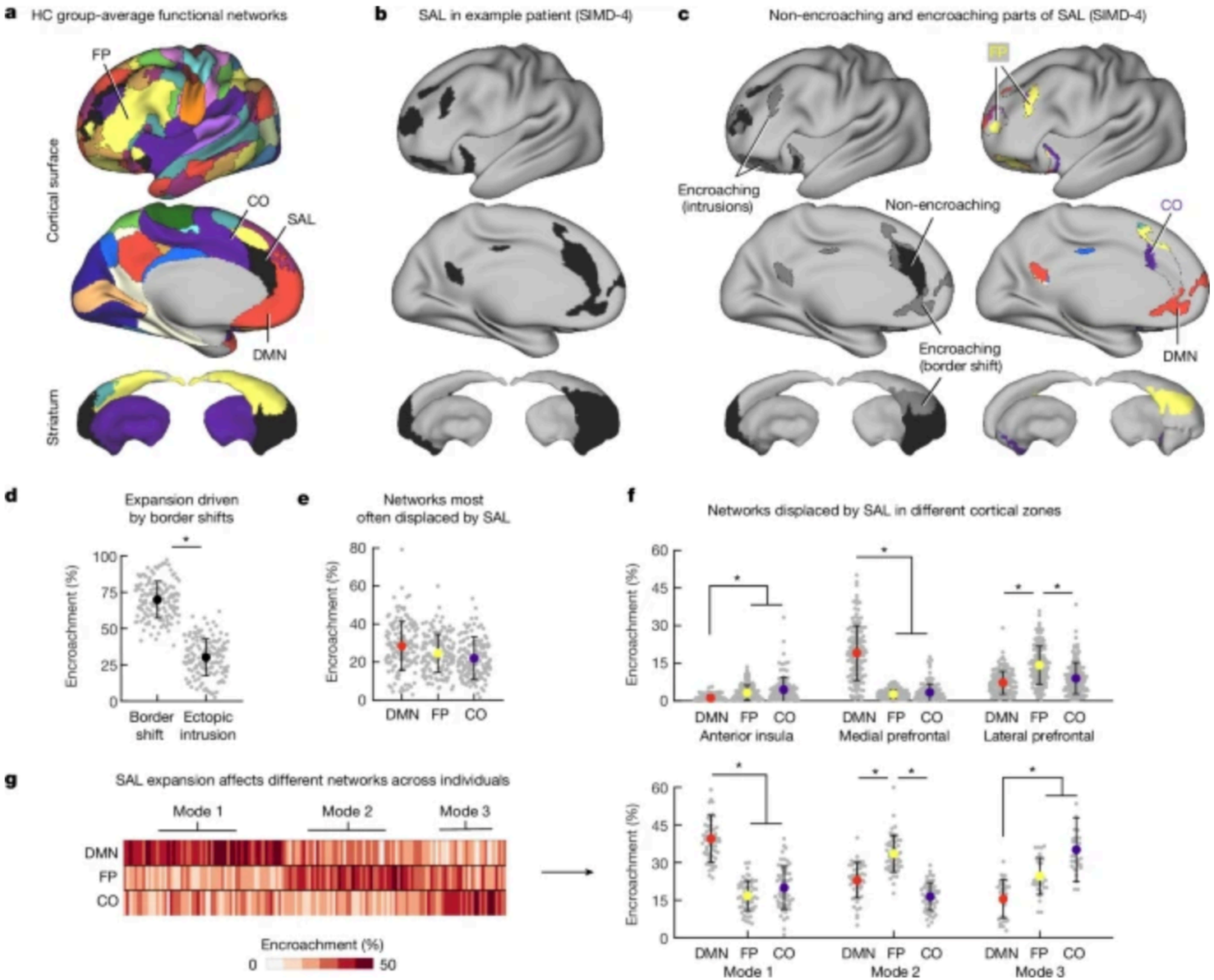
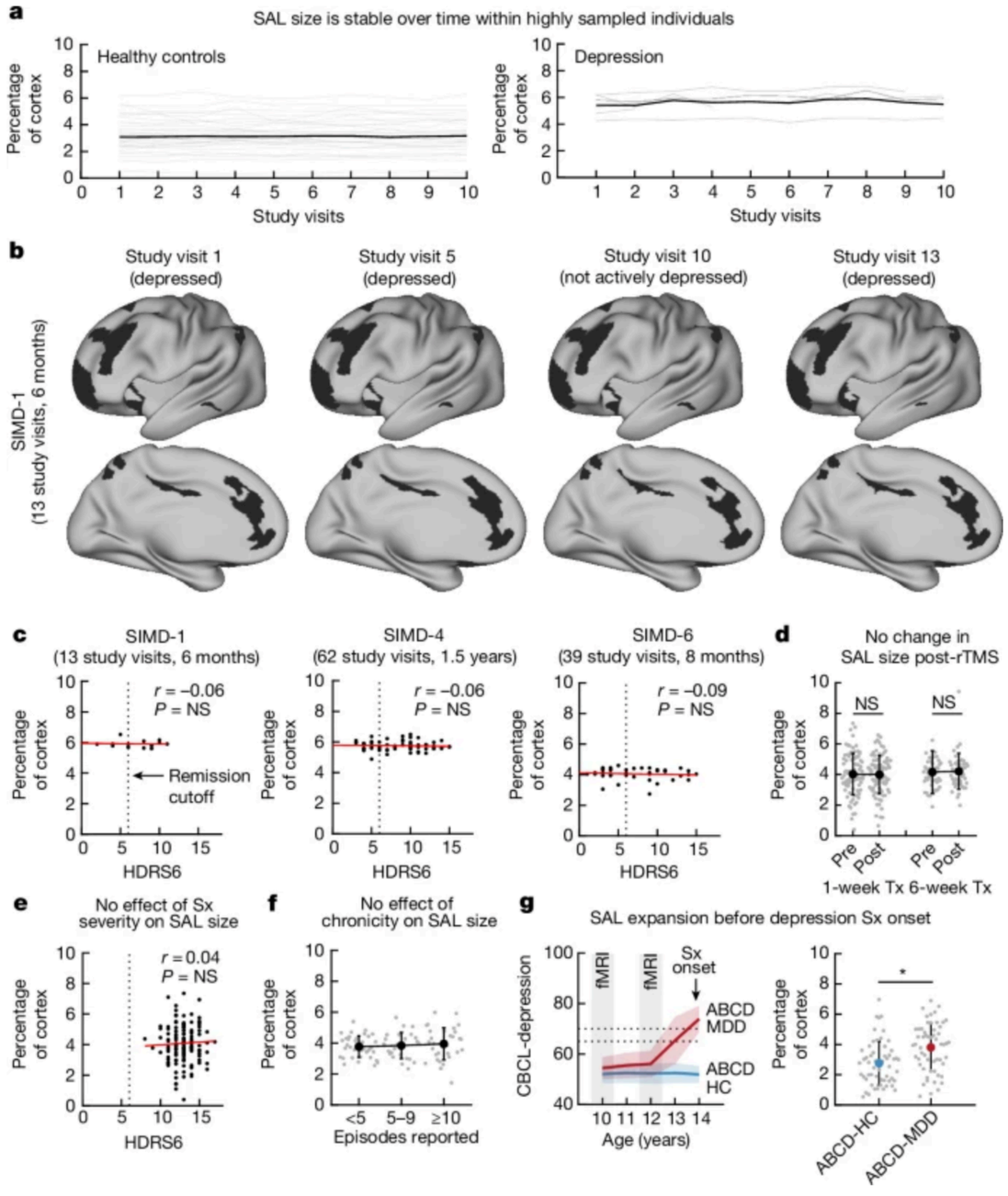
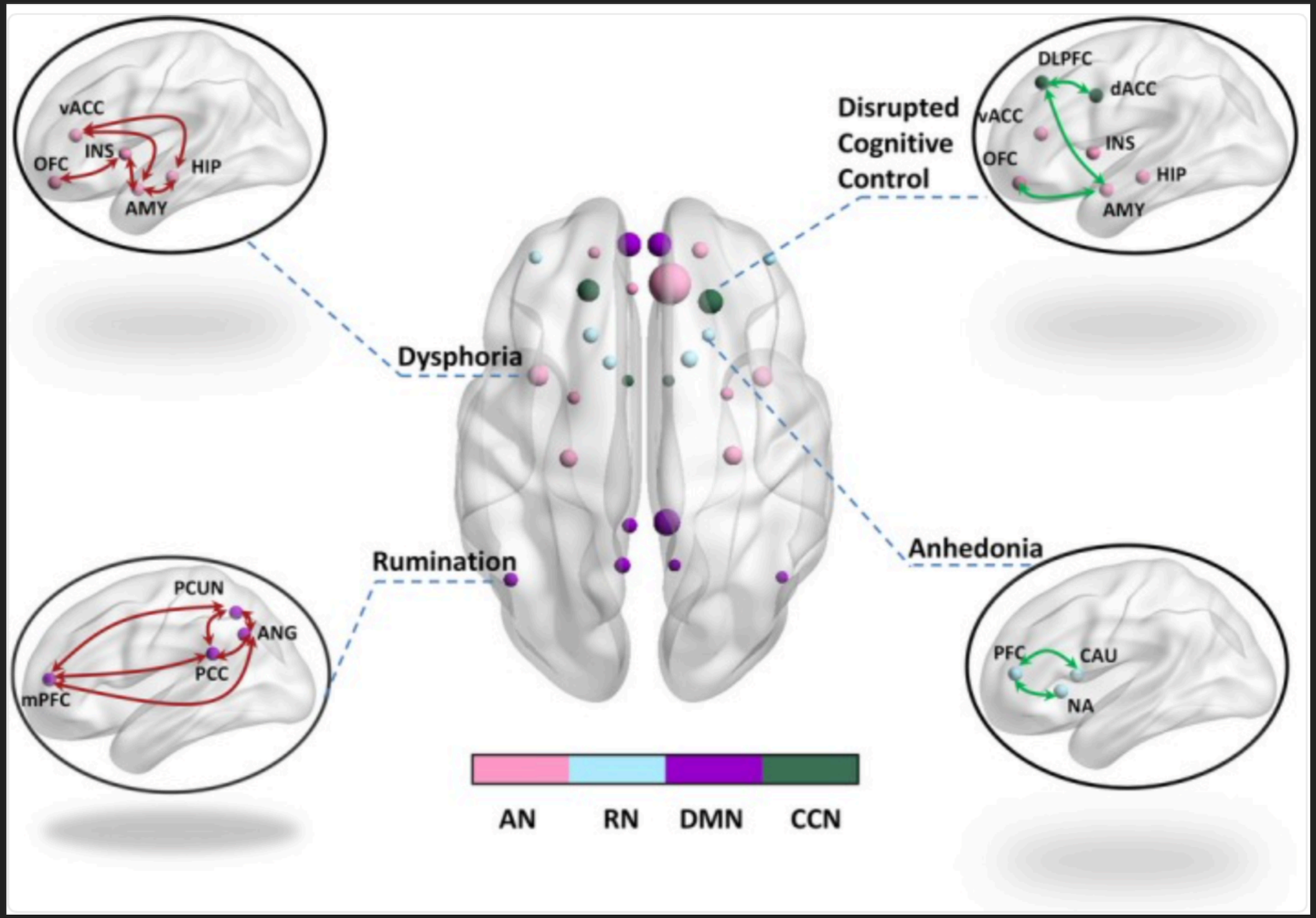


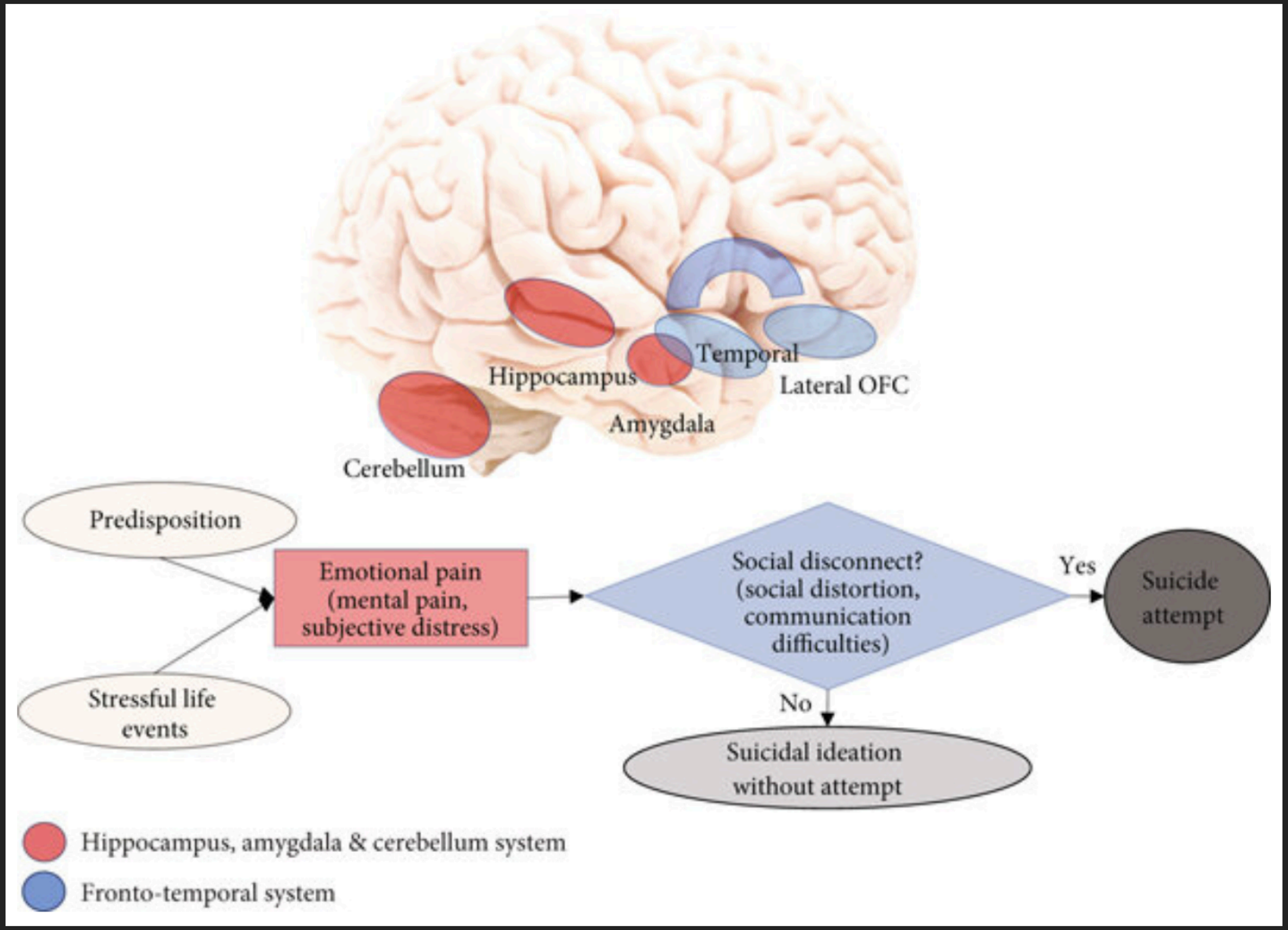
Fig. 3: Salience network expansion is stable over time and present before symptom onset.




vACC
INS
OFC
HIP
AMY
Dysphoria
DLPFC
dACC
Disrupted
Cognitive
Control
vACC
INS
OFC
HIP
AMY
Rumination
PCUN
ANG
PCC
mPFC
Anhedonia
PFC
CAU
NA
AN
RN
DMN
CCN

Neuroimaging Findings in Suicidality

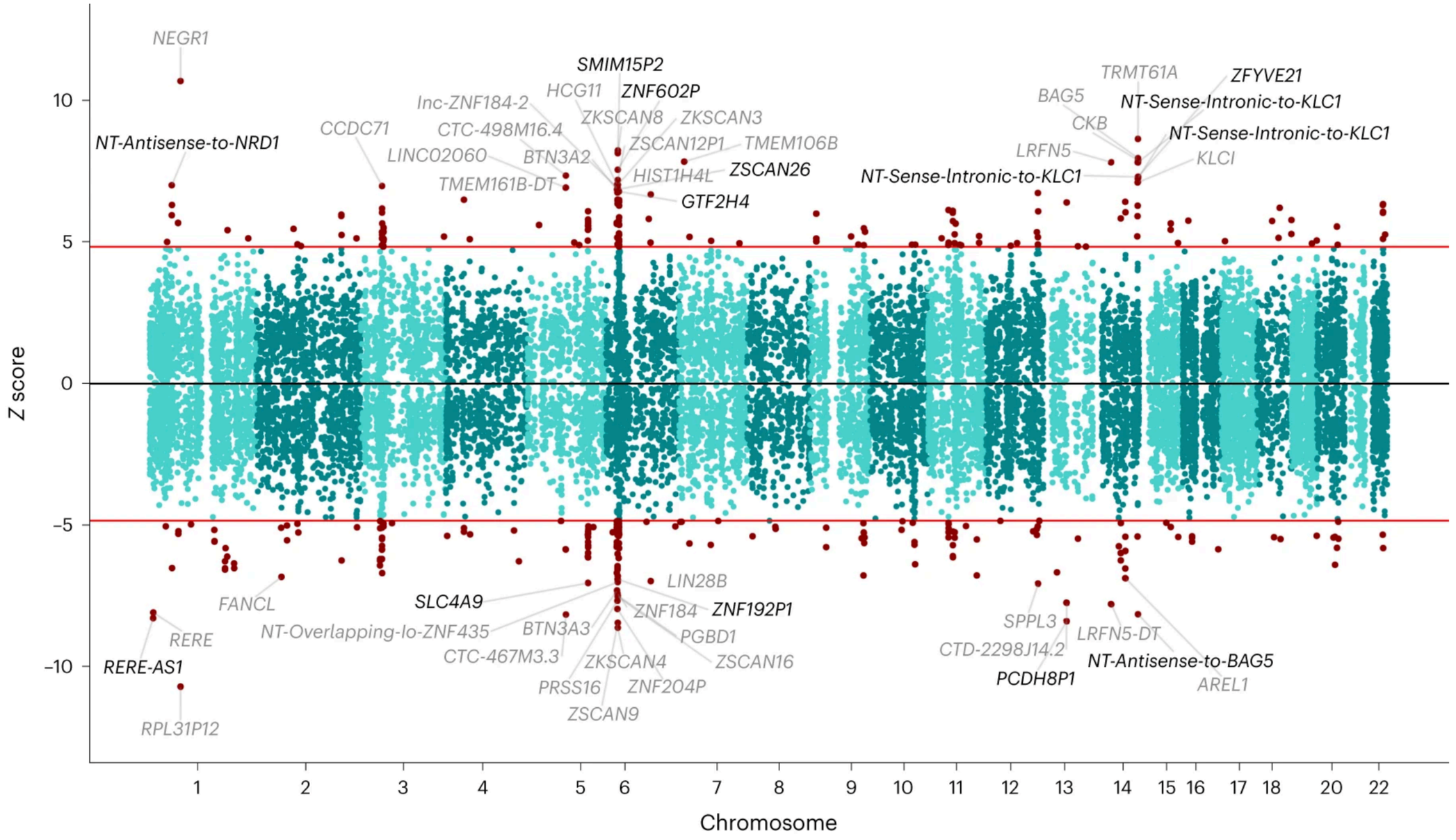
Neurobiological Perspectives


Temporal
Lateral OFC
Hippocampus
Amygdala
Cerebellum
Predisposition
Stressful life events
Emotional pain (mental pain, subjective distress)
Social disconnect? (social distortion, communication difficulties)
Yes
Suicide attempt
No
Suicidal ideation without attempt
Hippocampus, amygdala & cerebellum system
Fronto-temporal system

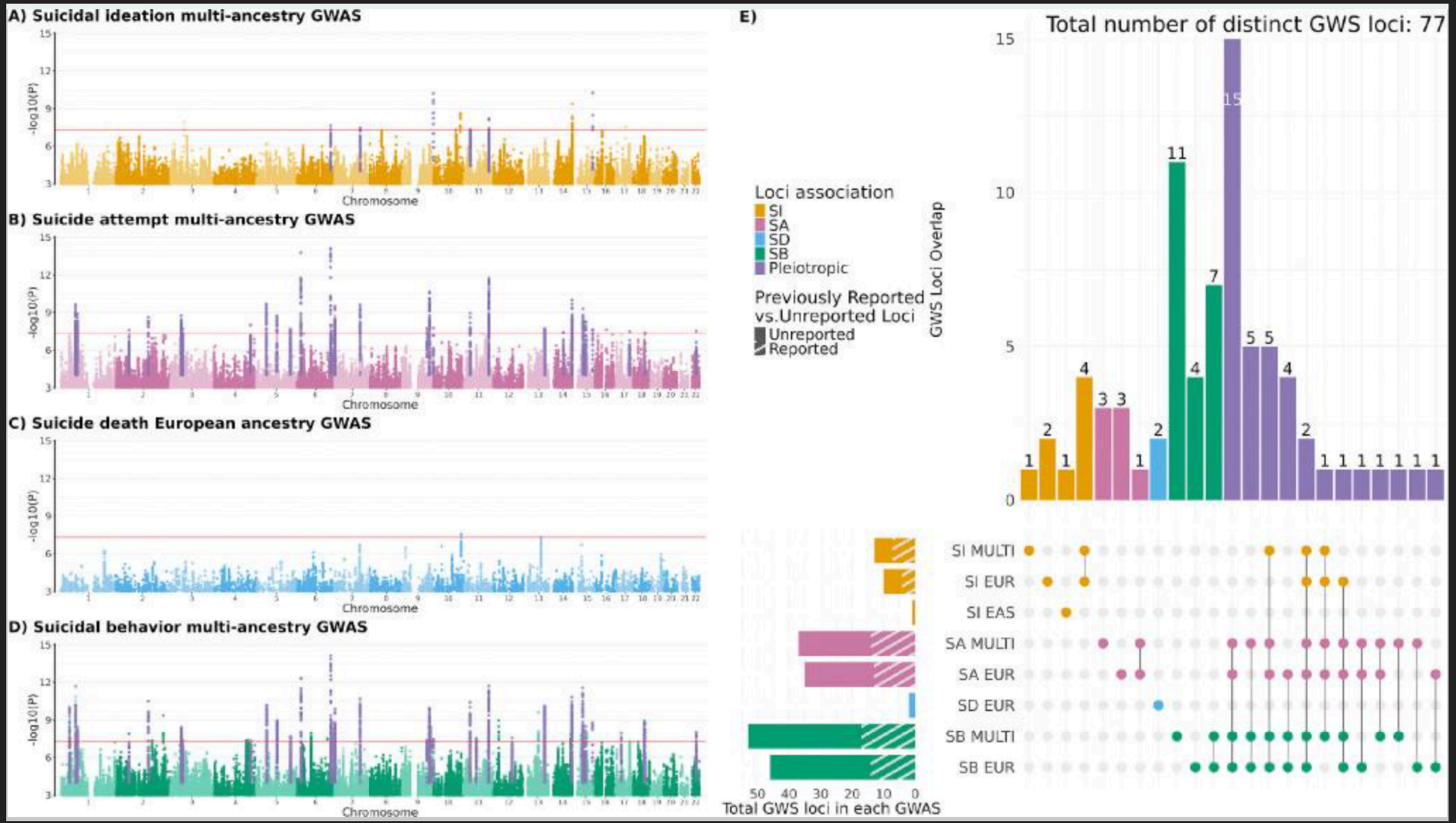
Genomic Findings

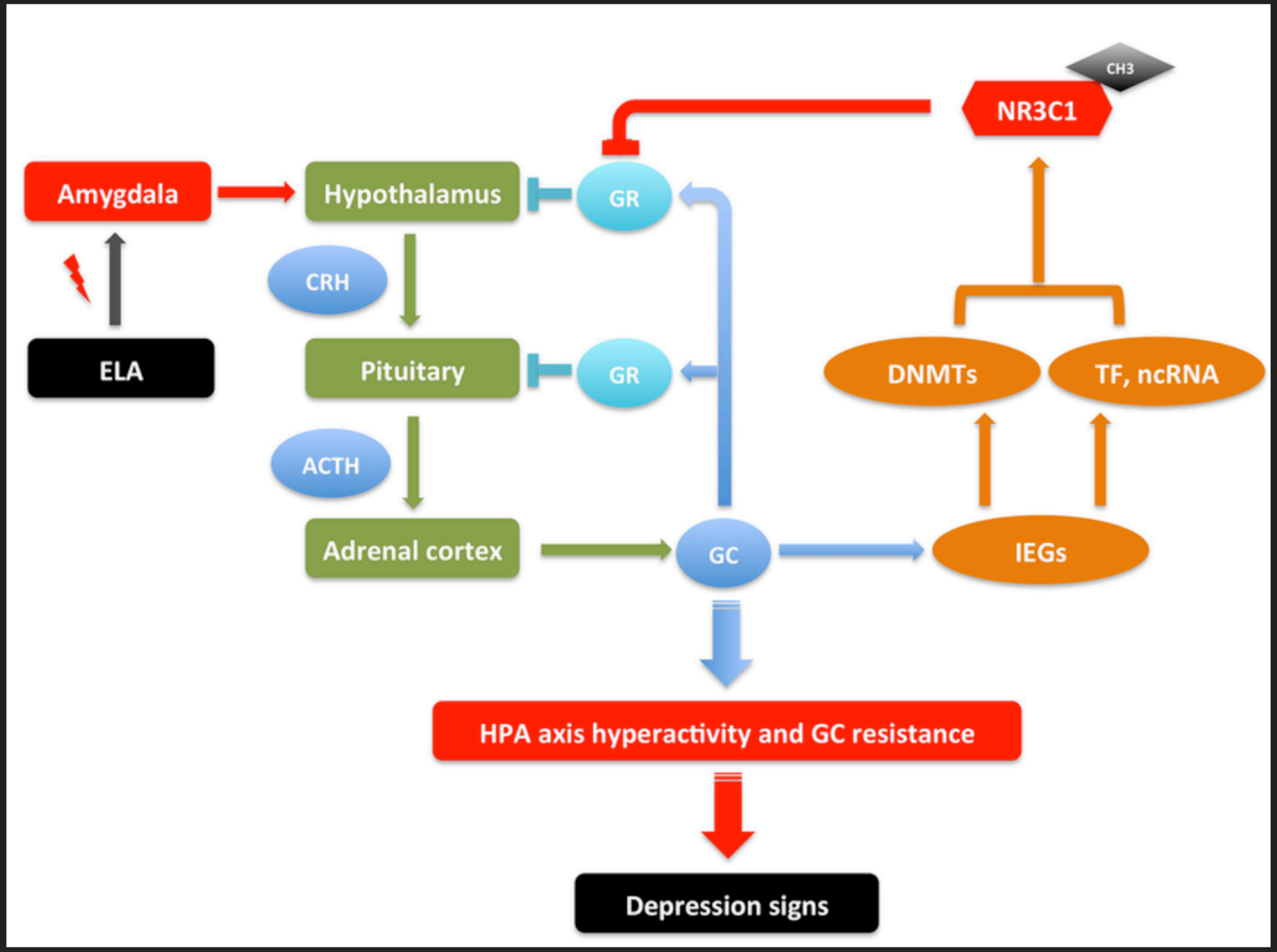
Neurobiological Perspectives

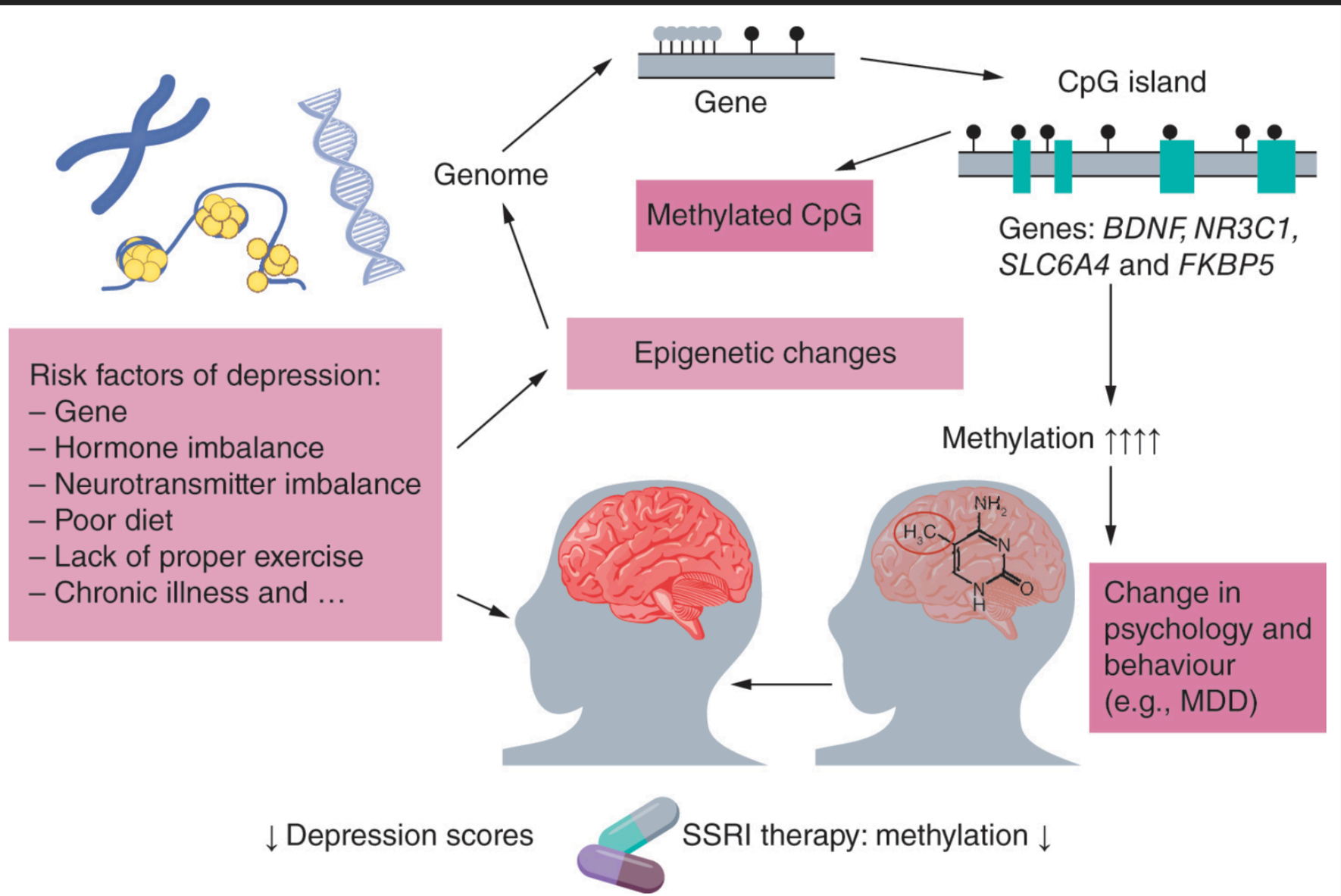
From: Multi-ancestry genome-wide association study of major depression aids locus discovery, fine mapping, gene prioritization and causal inference



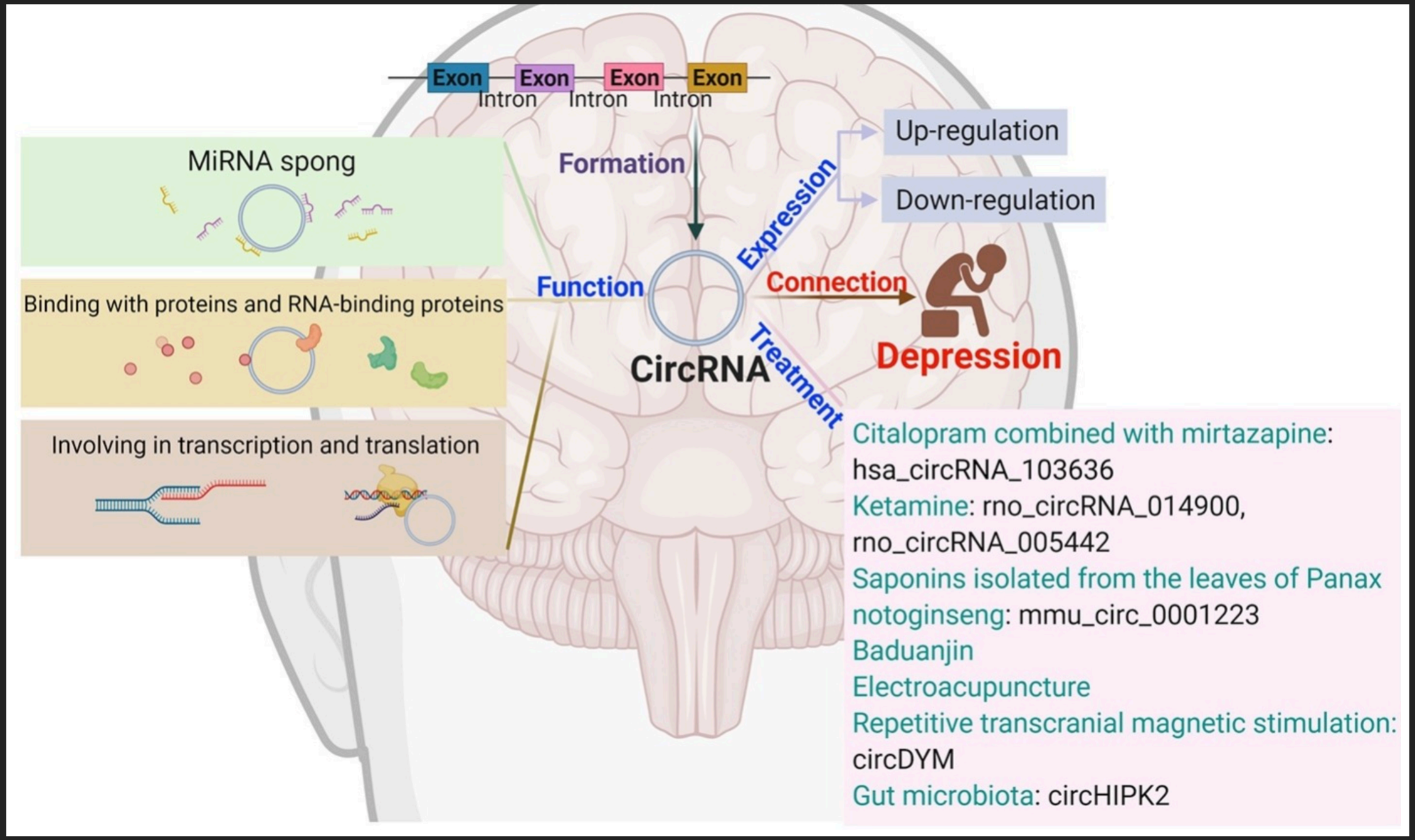
Significant gene associations are shown as red dots (354 significant genes, 205 of them novel), and the 50 most significant gene names are highlighted on


A) Suicidal ideation multi-ancestry GWAS
B) Suicide attempt multi-ancestry GWAS
C) Suicide death European ancestry GWAS
D) Suicidal behavior multi-ancestry GWAS
-log10(P)
Chromosome
E)
Total number of distinct GWS loci: 77
Loci association
SI
SA
SD
SB
Pleiotropic
Previously Reported vs.Unreported Loci
Unreported
Reported
GWS Loci Overlap
1 2 1 4 3 3 1 2 11 4 7 15 5 5 4 2 1 1 1 1 1 1
SI MULTI
SI EUR
SI EAS
SA MULTI
SA EUR
SD EUR
SB MULTI
SB EUR
50 40 30 20 10 0
Total GWS loci in each GWAS


CH3
NR3C1
Amygdala
Hypothalamus
GR
CRH
ELA
Pituitary
GR
DNMTs
TF, ncRNA
ACTH
Adrenal cortex
GC
IEGs
HPA axis hyperactivity and GC resistance
Depression signs



Overview of pathway connecting risk factors for depression to gene methylation.


Exon
Exon
Exon
Exon
Intron
Intron
Intron
MiRNA spong
Binding with proteins and RNA-binding proteins
Involving in transcription and translation
Formation
Function
CircRNA
Expression
Up-regulation
Down-regulation
Connection
Depression
Treatment
Citalopram combined with mirtazapine: hsa_circRNA_103636
Ketamine: rno_circRNA_014900, rno_circRNA_005442
Saponins isolated from the leaves of Panax notoginseng: mmu_circ_0001223
Baduanjin
Electroacupuncture
Repetitive transcranial magnetic stimulation: circDYM
Gut microbiota: circHIPK2

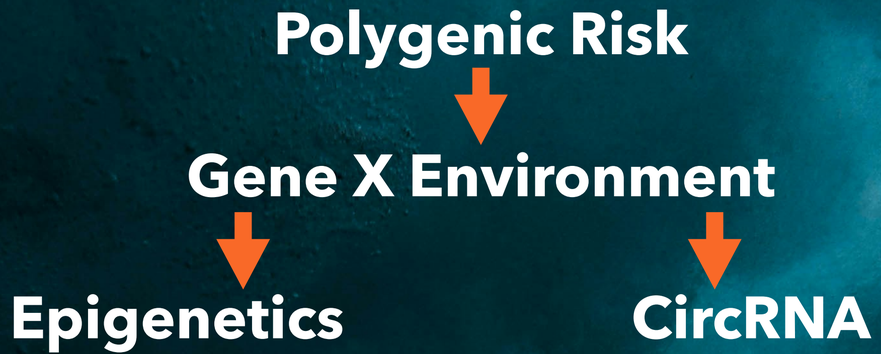
Multi-Level Model of Vulnerability

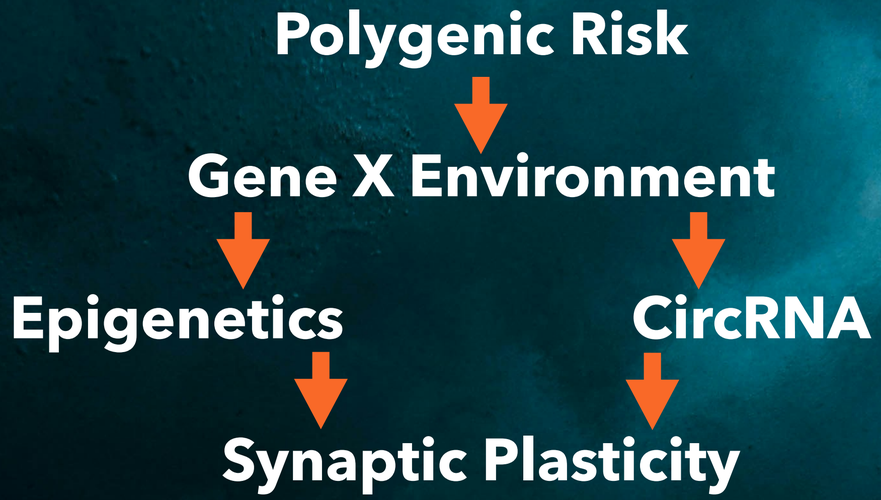
Polygenic Risk

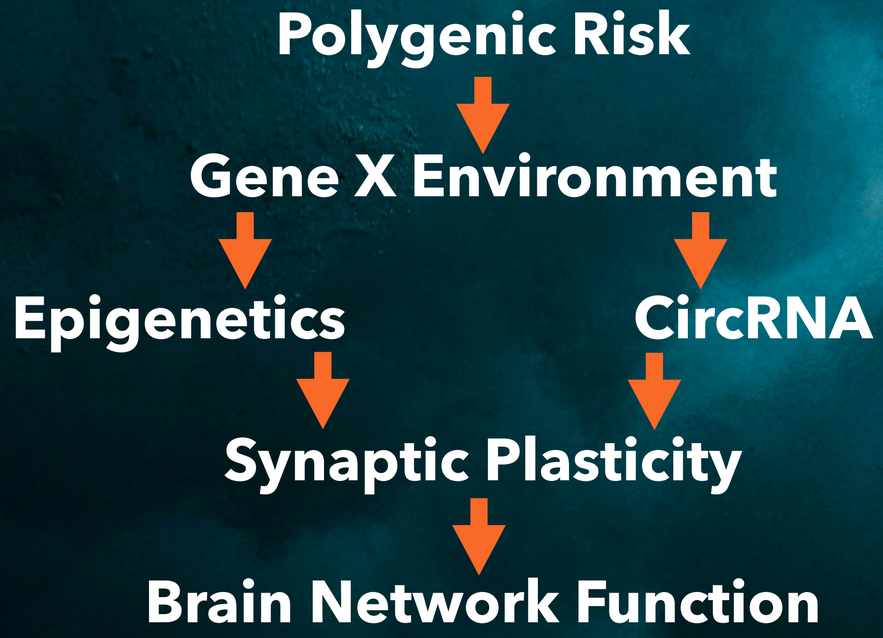
Polygenic Risk

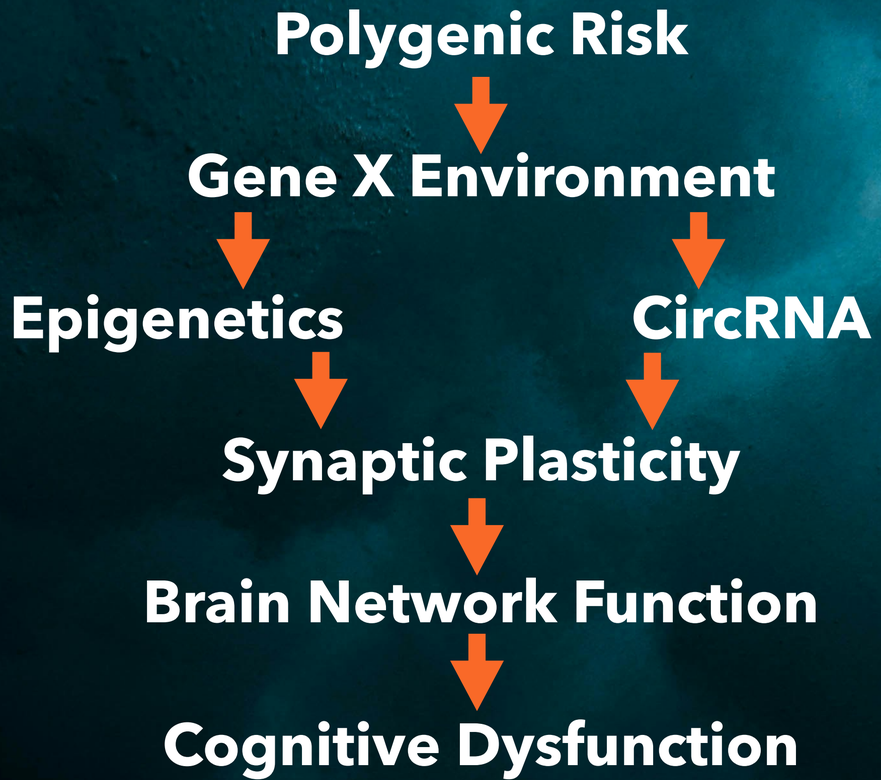
↓

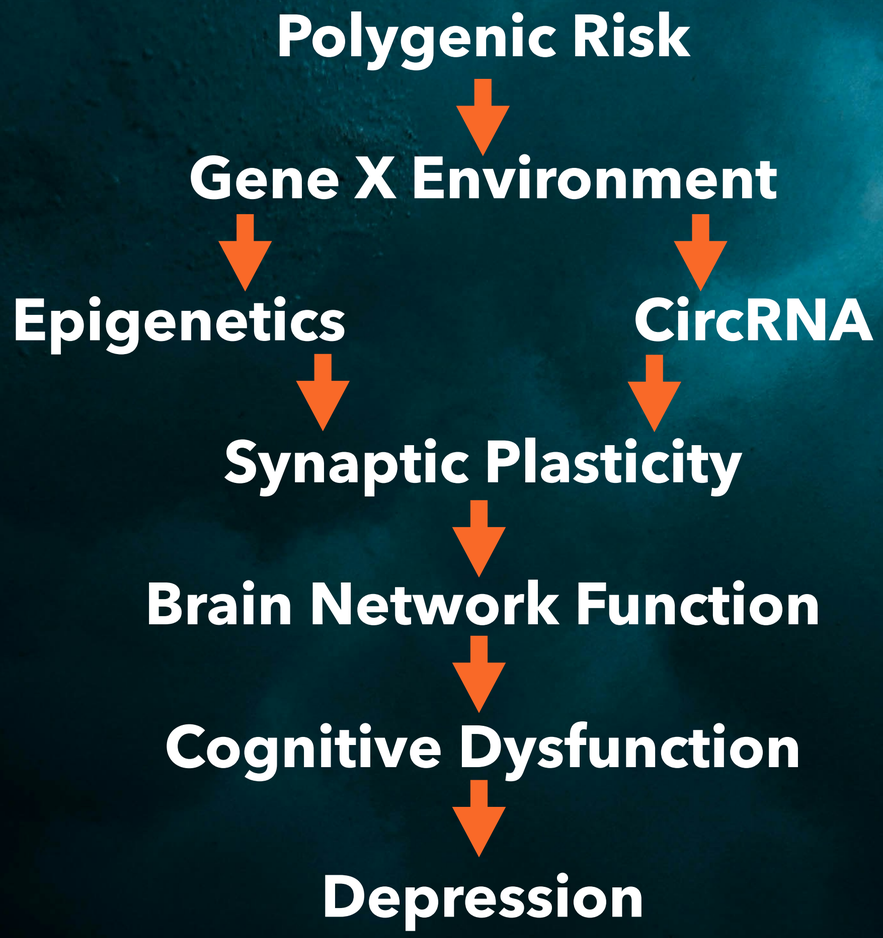
Gene X Environment

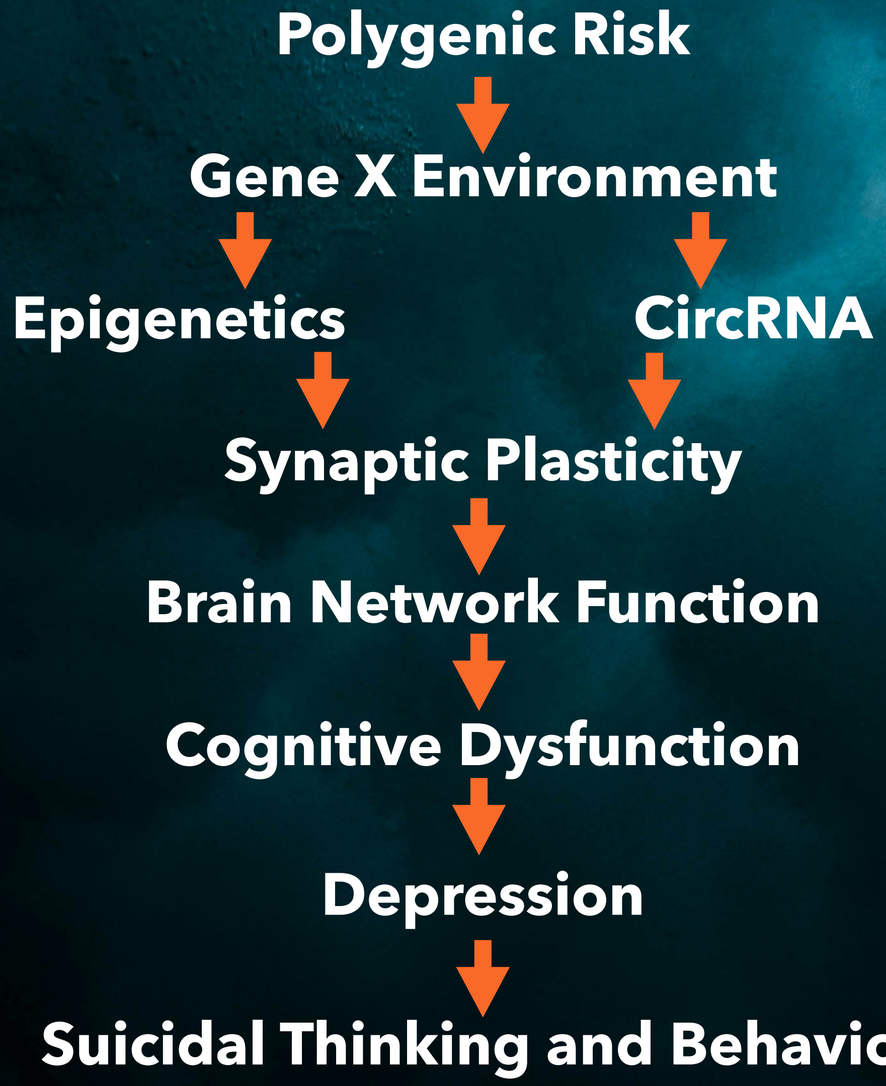

Polygenic Risk
Gene X Environment
Epigenetics
CircRNA


Polygenic Risk
Gene X Environment
Epigenetics
CircRNA
Synaptic Plasticity


Polygenic Risk
Gene X Environment
Epigenetics
CircRNA
Synaptic Plasticity
Brain Network Function


Polygenic Risk
Gene X Environment
Epigenetics
CircRNA
Synaptic Plasticity
Brain Network Function
Cognitive Dysfunction


Polygenic Risk
Gene X Environment
Epigenetics
CircRNA
Synaptic Plasticity
Brain Network Function
Cognitive Dysfunction
Depression


Polygenic Risk
Gene X Environment
Epigenetics
CircRNA
Synaptic Plasticity
Brain Network Function
Cognitive Dysfunction
Depression
Suicidal Thinking and Behavior

Take Home Messages

Depression is not a single disorder but an emergent property of interacting biological and psychological systems

Suicidal ideation and behavior arise from partially distinct mechanisms

Risk emerges through interactions between polygenic vulnerability, environmental stress, epigenetic regulation and brain network function

The future lies in integrated approaches that leverage genomics, neuroimaging, psychological theory and computational modeling to reveal mechanistic underpinnings

Reccomended Reading

1. Genetics & Molecular Mechanisms
   - Wray NR et al. (2018). GWAS of Major Depression.
   - Howard DM et al. (2019). Genome-wide meta-analysis of depression.
   - Nestler EJ et al. Epigenetic mechanisms of depression.
   - Klengel T & Binder EB. Gene-environment interactions and stress-related disorders.
   - Rybak-Wolf A et al. Circular RNAs in the mammalian brain.
   - You X et al. Neural circular RNAs are derived from synaptic genes and regulated by development.
- Depression Theory
   - Aaron T. Beck (1967). Depression: Clinical, Experimental, and Theoretical Aspects.
   - Denny Borsboom. A Network Theory of Mental Disorders.
- Suicide Theory
   - Thomas Joiner. Why People Die by Suicide.
   - Rory C O'Connor & Olivia J Kirtley. IMV Model.
- Evolution
   - Paul W Andrews & J Anderson Thomson Jr. Analytical Rumination Hypothesis.
- Neuroimaging
   - Matthew Dobbertin et al. (2023). Review of MRI findings in suicidality.
   - Nicola Meda et al. (2024). Neuroimaging meta-analysis of suicide attempts.
   - Xiaohan Zhang et al. (2024). Network localization of suicide-related brain abnormalities.